WORKBOOK 01 / 08 · MULTIMEDIA 12

Make a story the audience can read

Story beats, staging, scene construction and an animatic

THREE SCENES · ONE CAUSE-AND-EFFECT STORY

Discovery



Goal → decision

Crossing



Walk → jump → retrieve

Me_clever



Return cut → count → pride

Planning lengths: 144 f / 240 f / measured voice + ending hold

Three main Scenes form one cause-and-effect sequence. Lengths remain planning estimates until the voice is recorded.

Build: Build a readable three-Scene animatic before polishing any movement.

Keep: Caveman_01_story.fla + six-panel storyboard + silent animatic review

Shared story: A Caveman wants a third counting stone. A stream stands in his way. He studies it, commits to a jump, retrieves the stone and returns to count his collection with enormous pride.

Document: ActionScript 3.0 · 1280 × 720 · 24 fps · three main Scenes · Graphic symbols for scrubbable character actions

Two provisional 70-minute sessions. Use the HTML guide for read-aloud, switchable supports, recovery files and narrated explanation. This print version includes the essential support prompts.

Continue the same film

Create the project in Workbook 1. For every later workbook, open the selected previous production FLA and Save As a new numbered checkpoint, preserving all three Scenes and the complete Library. Keep repair/director copies separate; return to the chosen production version.

Understand before you animate

Staging directs attention. A story beat changes what the character knows, wants or does.

A Scene is a main-timeline section in Animate. A shot is one uninterrupted camera view. One Scene may contain several shots. This project begins with one shot per Scene so that story and file structure remain manageable.

Today’s checklist

1. Predict the result. 2. Build four stages. 3. Check in playback. 4. Refine, diagnose and direct. 5. Save and reopen.

Independent director condition

Move the third stone to a less obvious place. Change staging, gaze or timing so a viewer can still identify the goal without narration.

What you will defend

Where does the audience first understand the goal, and which visual choice makes it clear?

Support choices

Use the step summaries, glossary, numbered checklist, different-object examples, explanation frames and optional stretch. Open the companion HTML for browser read-aloud and the narrated six-slide explanation. Supports change access to the task; the success criteria remain the same.

Your planning notes

01 · Turn the premise into six changes

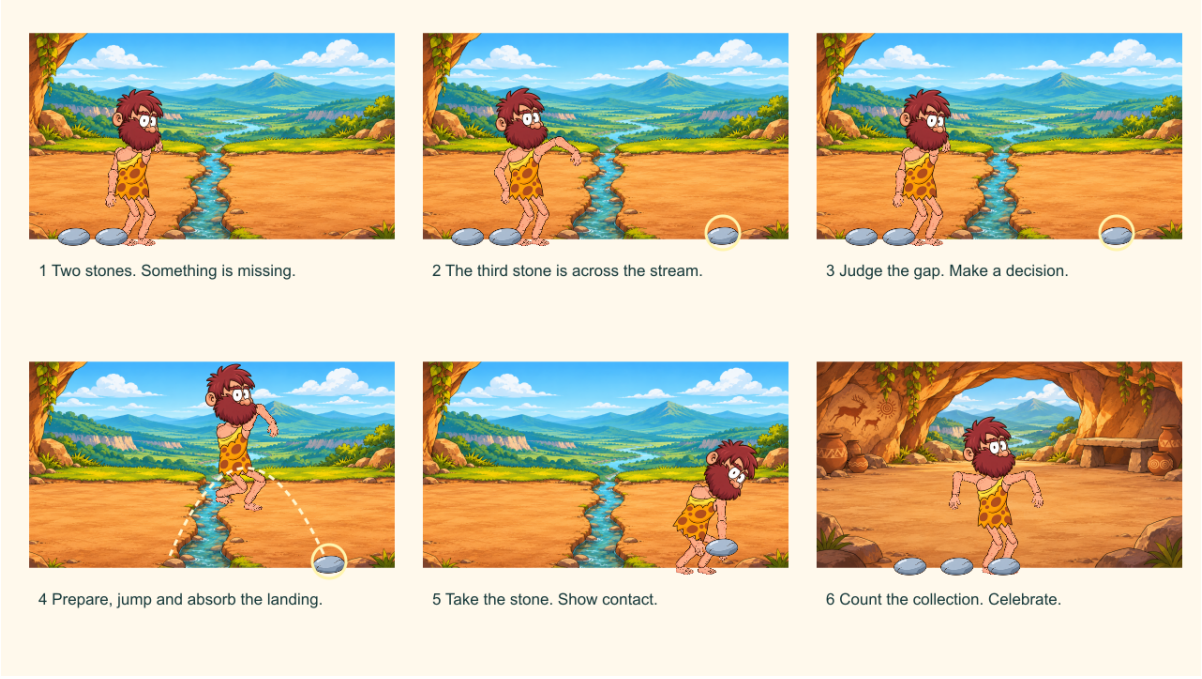
SELECT: Paper or the printable storyboard; no Animate selection yet.

TOOLS / ROUTE: Storyboard → goal → obstacle → decision → action → result.

1. Write six captions: notice two stones; spot the third; judge the stream; jump and land; retrieve the stone; count and celebrate.

2. Draw a thumbnail for each caption. Show the Caveman, the important prop and the direction of attention. Use arrows only for movement, not as a substitute for drawing the next pose.

3. Below each panel, write what changes. If two panels communicate the same information, combine them.



Illustrated story thumbnails: six changes develop one goal. These are authored planning drawings, not application screenshots.

WHY · Viewers need a chain of causes. A collection of effects does not explain why he jumps.

CHECK / SUMMARY · Cover the captions. A partner should be able to describe the want, obstacle and result from the drawings.

FIX · If the third stone looks like background decoration, isolate its silhouette, move it into the gaze line or give its reveal a short hold.

GLOSSARY · beat: A meaningful change in action, information or intention.; thumbnail: A small, quick composition drawing.

DIFFERENT-OBJECT EXAMPLE · A squirrel sees a nut behind a fence, tests the gap and chooses a route. Each panel changes its decision.

STRETCH · Add a single visual joke after the result, without introducing a second story.

02 · Create the film document

SELECT: New document preset; then click an empty area of the Stage.

TOOLS / ROUTE: File → New → ActionScript 3.0; Properties → Doc.

1. Create an ActionScript 3.0 document. Set width 1280, height 720 and frame rate 24 fps. Use pixels.

2. Save as Caveman_01_story.fla in a project folder containing Assets, Audio, Exports and Evidence.

3. Write the document type and frame rate on your storyboard. Keep 24 fps through the series.

WHY · The document type affects sound and camera workflows. A stable frame rate makes timing calculations meaningful.

CHECK / SUMMARY · Doc properties show 1280 × 720 and 24 fps; the file has a saved location.

FIX · If you created HTML5 Canvas, create the correct document now and import the artwork again. Do not assume that changing publish settings changes document type.

GLOSSARY · Stage: The visible movie area.; frame rate: The number of timeline frames played each second.

DIFFERENT-OBJECT EXAMPLE · At 24 fps a 48-frame shot lasts two seconds; at 12 fps the same frames last four.

STRETCH · Explain why increasing fps does not add new designed poses.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

03 · Build the three main Scenes

SELECT: The main timeline, not the inside of a symbol.

TOOLS / ROUTE: Window → Other Panels → Scene; Insert → Scene; double-click Scene names.

1. Rename Scene 1 Discovery. Add Crossing and Me_clever. Order them Discovery → Crossing → Me_clever in the Scene panel.

2. In each Scene create Background, Props, Character and Notes layers. Lock Background after placement; keep Notes as a guide layer so notes do not appear in the movie.

3. Return to Discovery using the Scene panel. Check the breadcrumb above the Stage; it must show the Scene, not a nested character symbol.

WHY · Scenes divide the production into story sections. Named layers reduce wrong-object edits.

CHECK / SUMMARY · All three names are visible in the Scene panel and play in story order.

FIX · If artwork disappears after switching Scenes, it may belong to another Scene. Import or paste a deliberate copy into the intended Scene; do not duplicate random Library items.

GLOSSARY · Scene: A section of the main movie timeline.; layer: A separate timeline track for artwork or sound.

DIFFERENT-OBJECT EXAMPLE · A three-shot weather report can use Intro, Forecast and Signoff Scenes with the same layer naming.

STRETCH · Plan when a shot change within a Scene would be clearer than adding another Scene.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

04 · Stage the goal and obstacle

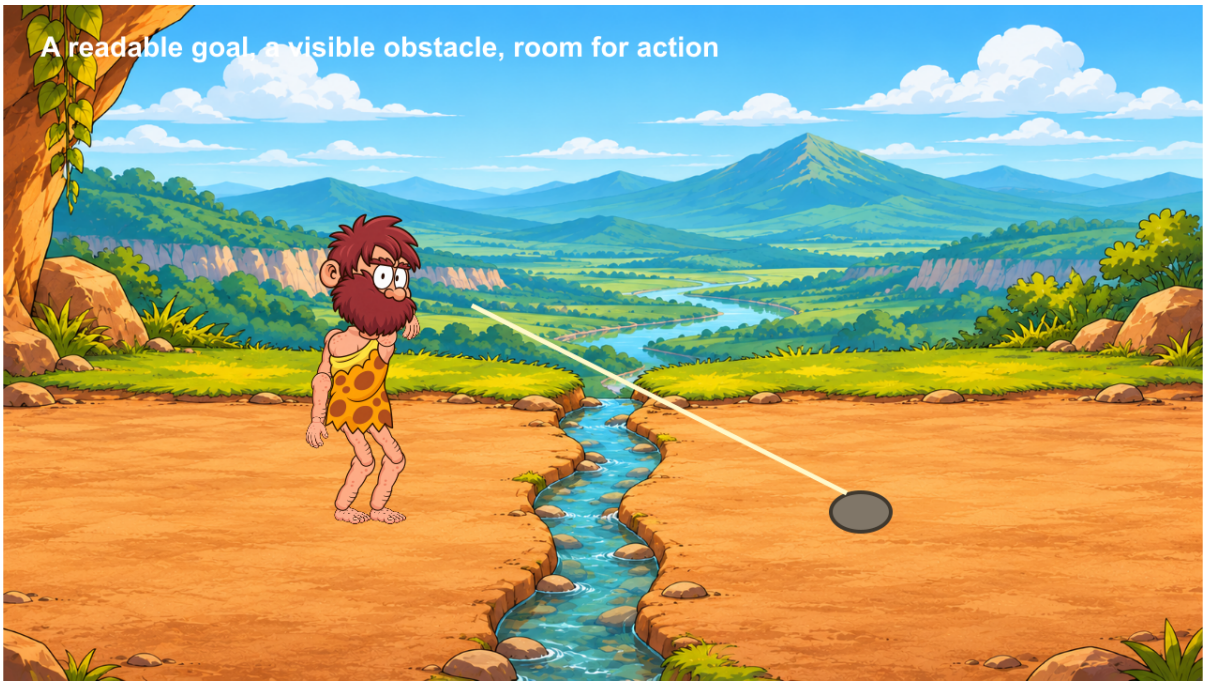
SELECT: Background layer frame 1; then the Props layer.

TOOLS / ROUTE: File → Import → Import to Stage; Properties → Position and Size; Modify → Convert to Symbol.

1. Place the stream background in Discovery and Crossing, and the cave background in Me_clever. This establishes the goal before the jump. Scale to fill the Stage without stretching one dimension independently.

2. Place two stones near the Caveman on the near bank in Discovery. Show the third stone clearly on the far bank in BOTH Discovery and Crossing. Keep its location consistent when the Scene changes. Convert each reusable prop to a Graphic symbol named Stone.

3. Keep a clear path of action from the near bank to the far bank. Use the Caveman reference as a temporary blocked pose. Do not animate the flattened reference as if its limbs were articulated.



Staging study: the stream is an obstacle and the isolated stone is a goal. Authored illustration using supplied Caveman art.

WHY · The viewer must read the destination and the danger before the jump begins.

CHECK / SUMMARY · At a small preview size, the Caveman, gap and goal are distinguishable. His feet can meet the ground plane.

FIX · Move the goal away from similarly coloured rocks. Enlarge it slightly for readability; do not add explanatory text as the first repair.

GLOSSARY · staging: Arranging elements to make an idea easy to read.; Graphic symbol: Reusable artwork with a timeline that can synchronize with its parent timeline.

DIFFERENT-OBJECT EXAMPLE · A bright envelope on a dark desk is easier to notice than one hidden among identical papers.

STRETCH · Compare two camera distances and justify which better explains the jump.

05 · Block the film with held poses

SELECT: Character layer frames in each main Scene.

TOOLS / ROUTE: Timeline → Insert Frame / Insert Keyframe; Onion Skin for reference.

1. Use held drawings or reference poses to block Discovery at about 144 frames and Crossing at about 240 frames. These are planning lengths, not final performance timings.

2. For Me_clever, reserve a temporary 144 frames. Label it temporary until the real recording is placed in Workbook 6.

3. Use a keyframe only when the pictured pose or composition changes. Add ordinary frames to extend a hold. Test the entire movie, not only the current Scene.

WHY · A held-pose animatic exposes pacing and missing story information before detailed animation becomes expensive.

CHECK / SUMMARY · The silent film shows a decision before the jump and a result after it. Scene changes occur once, in order.

FIX · If a hold is too short to understand, add frames. If everything feels slow, shorten the uninformative hold rather than deleting the decision.

GLOSSARY · animatic: A timed sequence of story drawings used to test a film.; hold: A pose sustained for several frames.

DIFFERENT-OBJECT EXAMPLE · A door creak can be blocked with closed, ajar and open drawings before animating the handle.

STRETCH · Show two versions of the decision hold and ask a viewer which implies greater caution.

06 · Use a deliberate cut back to the cave

SELECT: The end of Crossing and start of Me_clever.

TOOLS / ROUTE: Scene panel → Crossing / Me_clever; main timeline.

1. End Crossing with the stone visibly secured. Begin Me_clever with the third stone among the collection.

2. Keep the Caveman looking or moving in a compatible screen direction. Use the cave background to signal that time and location have changed.

3. Write one production note: “Return journey omitted; the cut advances time.” Do not accidentally imply that a single step teleports him across the stream.

WHY · Editing can omit routine travel while preserving cause and continuity.

CHECK / SUMMARY · A partner understands that he returned with the stone, rather than losing it or appearing in an unrelated place.

FIX · If the cut is confusing, add a brief departure pose or a settling pose at the cave. Add only the information the viewer needs.

GLOSSARY · ellipsis: A cut that omits a period of story time.; continuity: Consistency that helps an audience follow action across shots.

DIFFERENT-OBJECT EXAMPLE · A baker puts dough in an oven; a later shot shows bread cooling. The cut omits baking time.

STRETCH · Design a matched-action cut without increasing the number of Scenes.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

07 · Repair an unreadable version

SELECT: A duplicate saved as Caveman_01_repair.fla.

TOOLS / ROUTE: File → Save As; timeline holds and prop placement.

1. Deliberately make the goal stone hard to see or remove the decision hold in the duplicate. Play it to a partner without explaining.

2. Ask what the Caveman wants and why he jumps. Record the exact misunderstanding.

3. Repair one variable, replay and retain before/after frames. Explain why the repair addresses that misunderstanding.

WHY · Diagnosis connects visual choices to audience understanding.

CHECK / SUMMARY · Your evidence identifies a visible problem, one repair and a viewer response after the repair.

FIX · If your explanation says only “better,” name the missing information and the visual cue that now supplies it.

GLOSSARY · revision: A purposeful change based on a problem or feedback.

DIFFERENT-OBJECT EXAMPLE · Moving a button away from a busy pattern repairs visibility more directly than changing every page colour.

STRETCH · Test the film with the sound off and the screen reduced to a thumbnail.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

08 · Direct your first independent change

SELECT: Your own story file; preserve the guided version.

TOOLS / ROUTE: Save As → Caveman_01_director.fla.

1. Apply the changed-condition brief above. Keep the same goal, obstacle, three Scenes and character identity.

2. Re-stage the discovery; do not copy the worked composition unchanged. Test the silent animatic with a new viewer.

3. Save, close and reopen. Submit the storyboard, FLA and one before/after explanation using your teacher's named collection route. This local guide does not submit school work.

WHY · The independent variation shows whether you can transfer staging decisions.

CHECK / SUMMARY · The goal remains readable under your new condition and the reopened file retains all three Scenes.

FIX · If you changed the story to avoid the problem, restore the constraint and solve it through composition or timing.

GLOSSARY · transfer: Using a learned idea in a changed situation.

DIFFERENT-OBJECT EXAMPLE · Independent task: use the success criteria; no worked director solution is supplied.

STRETCH · Create a second ending only after the core three-Scene story reads clearly.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

Keep a small, useful evidence record

At frame ___, I chose ___ so the viewer could ___.
The problem was caused by ___. I changed ___ and checked ___.
Under the changed condition, I revised ___ because ___.

One repair

One independent decision and its trade-off

Delivery check

The selected file reopens. The result plays as intended. Joints/contact and audio were checked where relevant. The changed condition is solved. Required production files and a brief defence are retained.

Method references and attribution

A1 · Adobe Animate: symbols and instances · Adobe reference

A2 · Adobe Animate: timeline layers · Adobe reference

A3 · Adobe Animate: shape tweening · Adobe reference

A4 · Adobe Animate: parenting view · Adobe reference

A5 · Adobe Animate: sound · Adobe reference

A6 · Adobe Animate: camera · Adobe reference

Original teacher-supplied Caveman PDF/PNG assets are preserved. New matching Caveman parts/backgrounds are AI-generated and identified in the asset register. Teaching diagrams are authored illustrations; genuine native captures are labelled separately. Supplied commercial course videos/FLAs are reference material and are not redistributed.

Publication/classroom status: see the teacher companion. This file does not certify the independent cold run, a natural voice performance, or live assessment/progress connections.

WORKBOOK 02 / 08 · MULTIMEDIA 12

Build a Caveman you can pose

Parts, symbols, pivots, hierarchy and readable silhouettes

ACT THROUGH THE WHOLE BODY



Balanced Thinking Proud

Three full-body acting studies: balanced, thinking and proud.

Build: Assemble an articulated character and demonstrate intentional joint movement.

Keep: Caveman_02_character.fla + neutral, thinking and proud pose images

Shared story: A Caveman wants a third counting stone. A stream stands in his way. He studies it, commits to a jump, retrieves the stone and returns to count his collection with enormous pride.

Document: ActionScript 3.0 · 1280 × 720 · 24 fps · three main Scenes · Graphic symbols for scrubbable character actions

Two provisional 70-minute sessions. Use the HTML guide for read-aloud, switchable supports, recovery files and narrated explanation. This print version includes the essential support prompts.

Continue the same film

Create the project in Workbook 1. For every later workbook, open the selected previous production FLA and Save As a new numbered checkpoint, preserving all three Scenes and the complete Library. Keep repair/director copies separate; return to the chosen production version.

Understand before you animate

A clear silhouette communicates action before details. A joint pivot controls how a body part rotates.

A flattened PNG is a picture, not an articulated rig. Fourteen separate Graphic symbols allow posing. The supplied art uses five original component drawings, reused for separate arm and leg symbols. Reusing an image does not mean the symbol definitions are the same. Parenting is optional for this core manual-posing route.

Today’s checklist

1. Predict the result. 2. Build four stages. 3. Check in playback. 4. Refine, diagnose and direct. 5. Save and reopen.

Independent director condition

Pose the Caveman carrying the stone close to his chest. Preserve balance, clear hands and a grounded foot.

What you will defend

Show a joint pivot, explain why it belongs there, and identify one silhouette improvement.

Support choices

Use the step summaries, glossary, numbered checklist, different-object examples, explanation frames and optional stretch. Open the companion HTML for browser read-aloud and the narrated six-slide explanation. Supports change access to the task; the success criteria remain the same.

Your planning notes

01 · Choose and inspect the construction route

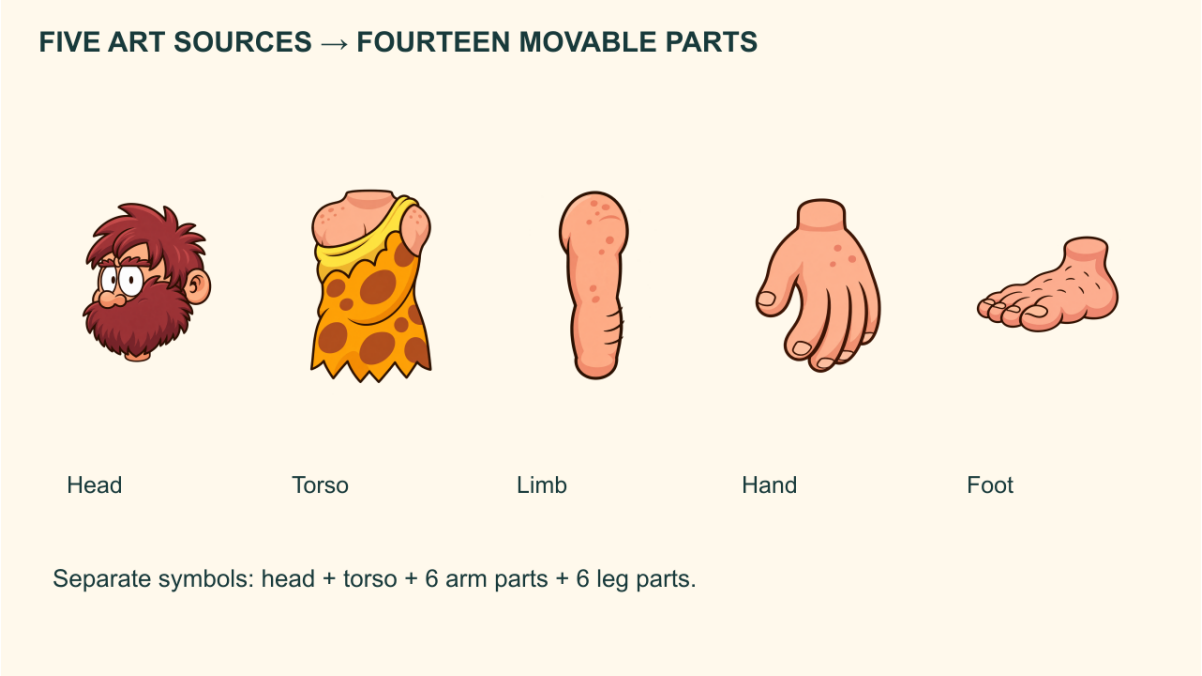
SELECT: Character artwork folder or the prepared character FLA.

TOOLS / ROUTE: File → Import → Import to Library; Window → Library.

1. Inspect the reference, head, torso, limb, hand and foot artwork. The original seated caveman_base.png is reserved for the counting close-up; it is not a walking body.

2. Choose the cutout route with supplied parts or draw your own matching parts with Brush/Pencil and fills. Both routes must create movable joints and the same story actions.

3. Use Far and Near to mean farther from and nearer to the viewer. Keep these names throughout the file.



Five original component artworks are reused in fourteen separate movable part definitions.

WHY · Construction choices affect what can move. Naming prevents the far and near limbs swapping identities halfway through a cycle.

CHECK / SUMMARY · You can identify the separate pieces and explain which picture is only a reference.

FIX · If you imported a single full-body image, retain it on a locked reference layer and build separate movable parts above it.

GLOSSARY · cutout animation: Animation using separately movable pieces.; silhouette: The readable outer shape of a pose.

DIFFERENT-OBJECT EXAMPLE · A paper bird needs a separate wing if the wing must flap independently.

STRETCH · Draw an alternate brow or hair tuft only if it can be placed without duplicating baked facial features.

02 · Create fourteen independent Graphic definitions

SELECT: One body part at a time; no background selected.

TOOLS / ROUTE: Modify → Convert to Symbol (F8); Window → Library → Duplicate.

1. Create Head and Torso. Create UpperArm_Far, Forearm_Far, Hand_Far, Thigh_Far, Shin_Far and Foot_Far; repeat the six names with _Near.

2. Use Graphic type. Duplicate a Library symbol before changing its internal artwork for another role. Give each stage instance its own layer of the same name.

3. When reusing the limb drawing, scale it inside each definition to suit upper arm, forearm, thigh or shin. Do not stretch the full character to fake bending.

WHY · Separate definitions let you revise one part without unintentionally replacing every limb.

CHECK / SUMMARY · Fourteen named Graphic definitions and fourteen part layers exist; selecting an arm does not select the whole Caveman.

FIX · If several pieces edit together, exit symbol editing and check whether you reused one definition. Duplicate and replace the appropriate instance.

GLOSSARY · definition: The reusable master stored in the Library.; instance: A placed use of a Library definition.

DIFFERENT-OBJECT EXAMPLE · Four wheel instances can share one wheel definition; a bent wheel needs a separate definition if only it should change.

STRETCH · Explain a situation where shared definitions are useful and one where they are dangerous.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

03 · Place pivots at the actual joints

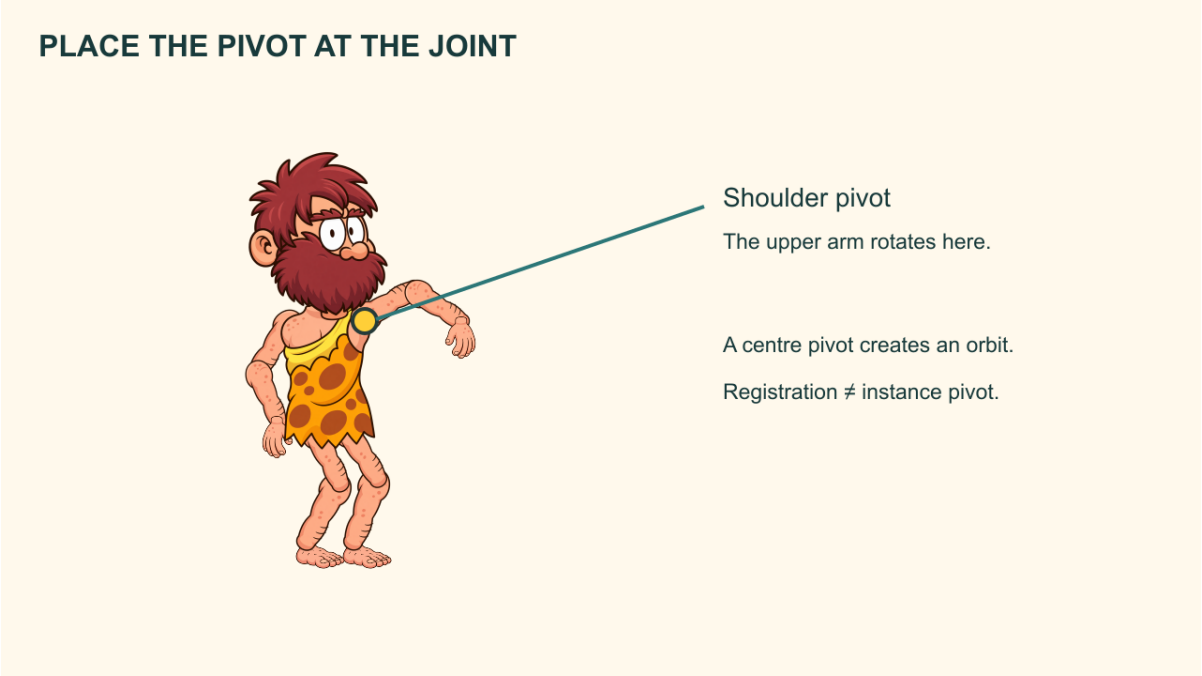
SELECT: An individual part instance with Free Transform (Q).

TOOLS / ROUTE: Free Transform → move the small white transformation point.

1. For each upper arm and thigh, move the transformation point to the shoulder or hip. Put the forearm pivot at the elbow, shin pivot at the knee, hand pivot at the wrist and foot pivot at the ankle.

2. Rotate a small amount, then undo. The joined end should remain in place. Test in both directions.

3. Distinguish the movable instance pivot from the symbol registration origin and from transparent PNG margins. Judge the joint by visible anatomy.



Joint-pivot mechanism diagram. The yellow point marks the shoulder; this is not a simulated Animate interface.

WHY · A centre pivot makes an arm orbit the shoulder. A joint pivot makes it rotate from the shoulder.

CHECK / SUMMARY · The test rotation leaves the joint attached. You can point to the pivot on the selected instance.

FIX · If the part swings away, undo and move the white point to the joint. Do not repair a pivot error by dragging the whole body after every rotation.

GLOSSARY · pivot: The transformation point used for an instance rotation.; registration: The symbol's internal coordinate origin.

DIFFERENT-OBJECT EXAMPLE · A door rotates from its hinge, not from the middle of the door panel.

STRETCH · Compare a hand waving from its wrist with an upper arm swinging from its shoulder.

04 · Assemble a neutral pose in depth order

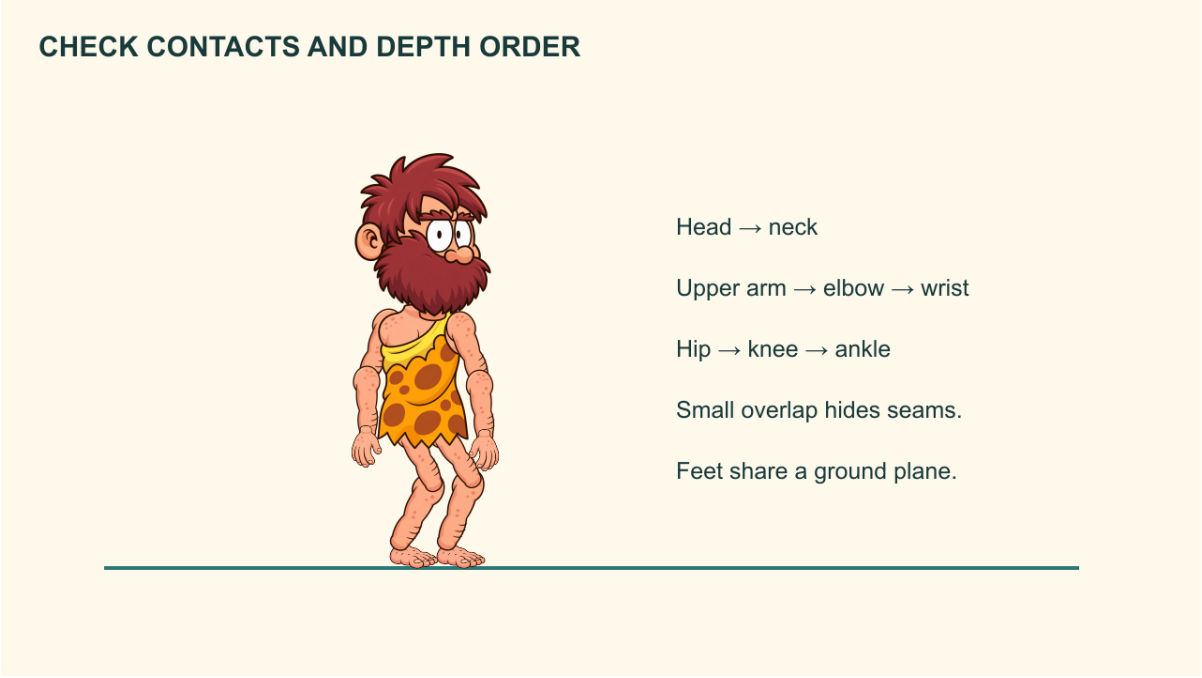
SELECT: The fourteen part layers at frame 1.

TOOLS / ROUTE: Timeline layer order; Selection and Free Transform.

1. Place feet on one ground line. Build shins to knees, thighs to hips, torso to hips, arms to shoulders and head to neck. Allow small joint overlaps so gaps do not open when posing.

2. Put far limbs behind the torso and near limbs in front where the pose requires it. Keep the head above the torso.

3. Select the complete assembly and convert it to a Graphic named Caveman_Action. Enter it, select the fourteen instances and use Modify → Timeline → Distribute to Layers if conversion put them on one layer; rename the resulting inner layers. Verify one part per inner layer, then inspect and restore the intended depth order; automatic distribution can follow creation order. Keep a neutral one-frame Library template.



Assembly study: connected joints, deliberate depth order and a shared ground line.

WHY · Layer order creates depth; overlap hides seams. A single action container lets the whole performance be staged later.

CHECK / SUMMARY · The neutral character has attached joints, a consistent head size and no floating feet.

FIX · If a limb disappears, check visibility, layer order and scale before importing another copy. If a dark halo appears, inspect the source edge on a light background.

GLOSSARY · overlap: Artwork covering a small part of another piece to hide a joint seam.; depth order: Which pieces appear in front of others.

DIFFERENT-OBJECT EXAMPLE · A puppet's sleeve overlaps its upper arm so the seam stays hidden while it bends.

STRETCH · Try a three-quarter pose without swapping the names Near and Far.

05 · Test a hierarchy without depending on it

SELECT: A duplicate test symbol, not the master neutral pose.

TOOLS / ROUTE: Timeline → Parenting view (Advanced Layers); or use the manual fallback.

1. Optional: enable the Parenting view, then parent a forearm layer to its upper arm and a hand to its forearm. Use the parent control or drag between parent circles as shown by your installed workspace.

2. Rotate the upper arm. Confirm whether both children follow. Undo and save the successful test separately. Parenting has not been certified by the supplied author test; stop this branch if the installed control differs.

3. Core fallback: leave parenting off. Rotate the upper arm, move the elbow end into place, rotate the forearm about its elbow, then place the hand at the wrist. Repeat for each designed pose.

WHY · A hierarchy can carry child transforms, but it does not choose a good pose, bend a knee intelligently or lock a foot to the ground.

CHECK / SUMMARY · Either you demonstrate the child-follow test or clearly identify that your file uses manual joint placement.

FIX · If a joint jumps after parenting, undo. Inspect pivots and the parent assignment in the duplicate; continue manually rather than breaking the master.

GLOSSARY · parenting: A relationship in which a child follows transformations of its parent.; inverse kinematics: A solving method that can calculate joint positions from an endpoint; ordinary parenting is not this.

DIFFERENT-OBJECT EXAMPLE · Moving a desk can carry a lamp placed on it, but does not automatically keep the lamp level in an animation hierarchy.

STRETCH · Build a one-chain parent test and show precisely what follows and what does not.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

06 · Design three readable acting poses

SELECT: Neutral template duplicated into a Pose_Test Graphic.

TOOLS / ROUTE: Insert Keyframe at frames 1, 13 and 25; Free Transform.

1. At frame 1 make a balanced neutral pose. At frame 13 make a thinking pose: head toward the goal, one hand near the beard, shoulders and hips still plausible.

2. At frame 25 make a proud pose: open chest, a deliberate hand gesture and a stable base. Rotate parts, reposition joints and inspect the line of action.

3. View the poses as silhouettes or squint at the Stage. Separate the hands from the torso when their meaning must be read.

WHY · Acting is communicated by the relationship of body parts, not just by a different mouth image.

CHECK / SUMMARY · A viewer can identify thinking and pride before you name the poses.

FIX · If two poses read the same, change the torso lean, head direction and negative space before adding small facial details.

GLOSSARY · line of action: A simple directional curve expressing the pose’s energy.; negative space: The clear spaces between parts of a silhouette.

DIFFERENT-OBJECT EXAMPLE · A person offering a gift extends the object into clear space; hiding it against the chest changes the message.

STRETCH · Make a doubtful proud pose by changing posture while keeping the same facial artwork.

07 · Repair the broken-joint example

SELECT: A copy of Pose_Test with one intentionally misplaced pivot.

TOOLS / ROUTE: Free Transform; timeline keys; Undo.

1. Move one elbow pivot to the centre of the forearm in the duplicate and rotate it. Capture the visible gap.

2. Restore the pivot at the elbow and reposition the wrist. Test the full range used by the thinking pose.

3. Save before/after evidence and explain the difference between the joint error and a layer-order error.

WHY · A useful repair demonstrates the cause of a failure, rather than hiding it behind another part.

CHECK / SUMMARY · The same rotation produces an attached joint after your repair.

FIX · If the gap remains, check that the pivot is correct on that keyframe instance and that the two visible joint ends overlap.

GLOSSARY · diagnosis: Identifying the cause of a visible problem.

DIFFERENT-OBJECT EXAMPLE · A wheel orbiting its axle is a pivot error; a wheel hidden behind the wrong panel is a depth-order error.

STRETCH · Check both the most bent and most extended poses for seams.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

08 · Pose the changed carrying condition

SELECT: A duplicate character action for your director version.

TOOLS / ROUTE: Save As; pose keys; Stage preview.

1. Create a carrying pose with the stone close to the chest. Use both hands if necessary, but keep the object visible.

2. Keep the supporting foot under the body's apparent weight. Show a front/three-quarter readability check at small size.

3. Save and reopen. Retain a neutral template and your three acting poses for the following workbooks.

WHY · A changed prop condition requires new joint relationships and balance decisions.

CHECK / SUMMARY · The stone appears held, hands remain attached and the pose differs intentionally from the guided examples.

FIX · If the stone floats, align fingers or palms around it and hold its position relative to the carrying hand.

GLOSSARY · balance: A pose's apparent ability to support its own weight.

DIFFERENT-OBJECT EXAMPLE · No worked carrying solution: use contact, silhouette and balance criteria.

STRETCH · Design a one-handed carry and explain the counterbalance in the torso.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

Keep a small, useful evidence record

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The problem was caused by ___. I changed ___ and checked ___.
Under the changed condition, I revised ___ because ___.

One repair

One independent decision and its trade-off

Delivery check

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Method references and attribution

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A2 · Adobe Animate: timeline layers · Adobe reference

A3 · Adobe Animate: shape tweening · Adobe reference

A4 · Adobe Animate: parenting view · Adobe reference

A5 · Adobe Animate: sound · Adobe reference

A6 · Adobe Animate: camera · Adobe reference

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WORKBOOK 03 / 08 · MULTIMEDIA 12

Make the Caveman think, then point

Key poses, designed breakdowns, in-betweens, timing and spacing

THE BREAKDOWN DESIGNS THE ROUTE



13 · thought



19 · hand clears beard



25 · point

A designed breakdown clears the hand around the face and lets the head lead.

Build: Build a short acting phrase whose transition is designed rather than left to default interpolation.

Keep: Think_Point Graphic + stepped and refined versions + a timing comparison

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Continue the same film

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Understand before you animate

Keys state the important poses. A breakdown designs how the action travels between them. In-betweens complete that designed transition.

A timeline keyframe stores a change; a storytelling key pose is an important idea in the performance. They are not synonyms. Timing is how long an action takes. Spacing is how positions are distributed through that time. A classic tween interpolates transforms; it does not understand anatomy, contact or intention.

Today’s checklist

1. Predict the result. 2. Build four stages. 3. Check in playback. 4. Refine, diagnose and direct. 5. Save and reopen.

Independent director condition

The Caveman notices a distant noise before pointing at the stone. Add a head-led change of attention without losing the original story goal.

What you will defend

Which drawing is your breakdown, and what does it decide that a mathematical midpoint would miss?

Support choices

Use the step summaries, glossary, numbered checklist, different-object examples, explanation frames and optional stretch. Open the companion HTML for browser read-aloud and the narrated six-slide explanation. Supports change access to the task; the success criteria remain the same.

Your planning notes

01 · Block the two storytelling keys

SELECT: A new Think_Point Graphic copied from the neutral character.

TOOLS / ROUTE: Library → Duplicate; symbol timeline; Insert Keyframe.

1. Keep neutral at frame 1. At frame 13 pose a clear thought: head inclines toward the beard hand. At frame 25 pose a clear point toward the goal.

2. Extend every part layer through frame 25. Keep only held keys while judging the performance; do not tween yet.

3. Write the intention under each key: “I am considering” and “There is my goal.”



Storytelling keys communicate consideration and a decision.

WHY · Keys must communicate the action before smoothness can improve it.

CHECK / SUMMARY · The two important poses have distinct silhouettes and the pointing hand indicates the goal.

FIX · If the point reads as a wave, straighten the intended arm direction and make the head/gaze support it.

GLOSSARY · key pose: An important pose that communicates the action or intention.; stepped blocking: Reviewing held poses before creating transitions.

DIFFERENT-OBJECT EXAMPLE · For a cat turning to a sound, the starting gaze and final gaze are keys; they do not yet specify the route.

STRETCH · Make the thought register using a torso shift as well as a hand position.

02 · Design the breakdown at frame 19

SELECT: All animated part layers at frame 19 inside Think_Point.

TOOLS / ROUTE: Insert Keyframe; Onion Skin; Free Transform.

1. Insert keys at frame 19 between the thought and point. Move the elbow outward so the hand clears the beard before extending.

2. Let the head begin to face the goal before the pointing arm fully extends. Keep shoulder, elbow and wrist connected.

3. Compare this pose with a simple halfway blend. Label the decision: “hand clears face; head leads.”

WHY · A breakdown chooses an arc, body relationship and order of movement. It need not be halfway in space or halfway in time.

CHECK / SUMMARY · The hand passes around the face rather than through it, and the viewer can follow the intended route.

FIX · If the hand cuts through the beard, widen the elbow path or move the breakdown earlier; do not add random frames around a bad route.

GLOSSARY · breakdown: A designed transition pose that explains how keys connect.; arc: A curved path followed by a moving point.

DIFFERENT-OBJECT EXAMPLE · A bird’s wing can fold near the body during recovery rather than taking the shortest straight-line route.

STRETCH · Place the breakdown at frame 17 instead and describe the change in emphasis.

03 · Add purposeful in-betweens

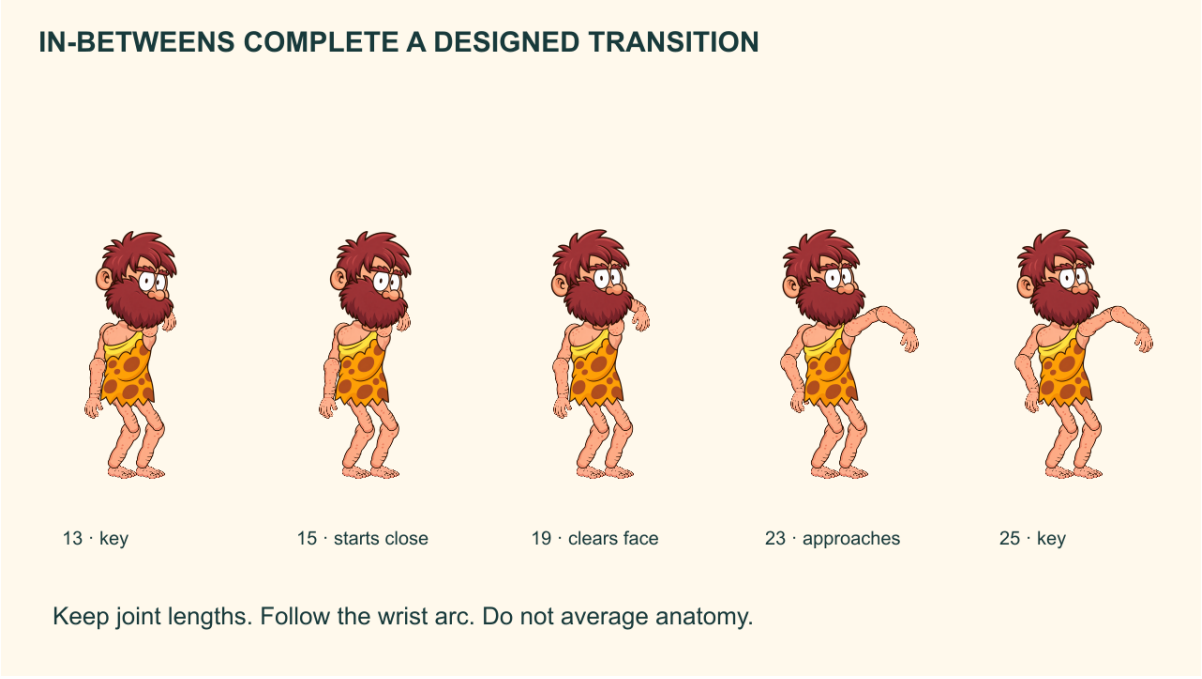
SELECT: Part layers between frames 13–19 and 19–25.

TOOLS / ROUTE: Insert Keyframe; Onion Skin; frame-by-frame posing.

1. Add in-between poses at frames 15, 17, 21 and 23. Keep the wrist on a smooth arc and maintain limb lengths.

2. Place the first movement close to the thought pose, allow larger spacing through the extension, then reduce spacing near the final point.

3. Play frames 13–25 repeatedly. Remove any in-between that introduces a kink; redraw or repose it instead of hiding it with more frames.



Transition planning strip. Additional poses follow a designed route while preserving joint lengths.

WHY · Additional drawings help only when their placement supports the intended motion.

CHECK / SUMMARY · The wrist path is smooth, joints remain connected and the final point does not snap unexpectedly.

FIX · If the forearm changes length, restore the correct elbow-to-wrist distance. If speed is uneven accidentally, compare adjacent wrist spacing.

GLOSSARY · in-between: A pose or drawing filling the transition between designed poses.; spacing: The distance between positions on successive frames.

DIFFERENT-OBJECT EXAMPLE · A pendulum has close positions near its turning points and wider positions through the bottom.

STRETCH · Try animating on twos, then add selected single-frame accents where the action needs precision.

04 · Compare timing while preserving the route

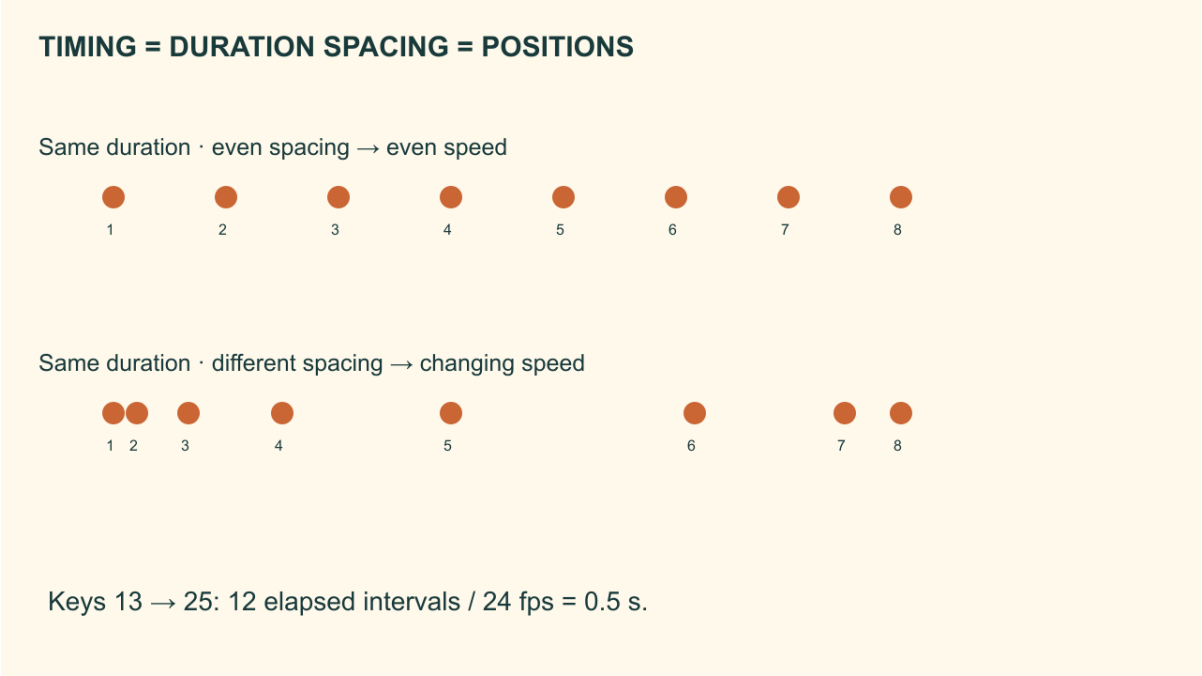
SELECT: A duplicate Think_Point_Slow; same poses and arc.

TOOLS / ROUTE: Timeline → move keyframes and extend all part layers.

1. Duplicate the action. In the slow version, move the final point from frame 25 to frame 37 and distribute the transition keys over the longer interval.

2. Keep the designed arc and relative pose order. Compare the two versions side by side or in sequence.

3. Calculate elapsed time between key 13 and key 25: 12 intervals ÷ 24 = 0.5 seconds. A 24-frame exported clip itself lasts one second.



Same frame intervals, different position spacing: duration and speed distribution are distinct.

WHY · Changing duration can change intention from quick recognition to careful consideration without changing the spatial route.

CHECK / SUMMARY · You distinguish interval duration from inclusive frame count and can explain the acting difference.

FIX · If a layer ends early, extend it to the new final frame. Do not change the document fps to make one action slower.

GLOSSARY · timing: The duration and placement of events in an action.; frame interval: The elapsed time from one frame position to the next.

DIFFERENT-OBJECT EXAMPLE · A lid closing over 6 intervals feels brisk; the same route over 24 intervals feels deliberate.

STRETCH · Keep duration fixed and change only spacing; describe why the result differs.

05 · Use a classic tween as a comparison

SELECT: A duplicate test action with one Graphic instance on each animated part layer.

TOOLS / ROUTE: Right-click between two keys → Create Classic Tween.

1. On the duplicate, test a classic tween between two pose keys. Select the span, not the Stage object, to inspect tween properties.

2. Scrub the midpoint and the elbow/hand contact. If a rotation takes the long way, inspect rotation settings and use the shortest suitable route.

3. Retain manual breakdown keys where the default interpolation produces a poor arc or broken joint. Do not claim the default tween designed your performance.

WHY · A tween can save repetitive placement, but it must be judged against pose and path decisions.

CHECK / SUMMARY · You can show an interpolated midpoint, identify a problem or justify why it works, and preserve your intended breakdown.

FIX · A dashed or failed span usually means unsuitable contents or mismatched symbols. Use one appropriate instance per layer and matching definitions at the ends.

GLOSSARY · classic tween: Interpolation of properties of an instance between keyframes.; interpolation: Calculating intermediate values between defined values.

DIFFERENT-OBJECT EXAMPLE · A sliding title is a good interpolation task; a hand avoiding a face needs an intentionally designed route.

STRETCH · Test uniform versus eased timing on the pointing hand and justify the selected version.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

06 · Add anticipation and a restrained settle

SELECT: Think_Point final pose and following frames.

TOOLS / ROUTE: Insert Keyframe at 11, 25, 28 and 32 as needed.

1. Extend EVERY part layer through frame 32 and extend the placed action's parent exposure to match. Before the point, let the hand draw slightly inward and the torso prepare. Keep this anticipation small enough to preserve the thought.

2. Let the arm reach a little beyond its settled point at frame 25, recover at 28 and rest at 32. Repose joints; do not merely wobble the whole picture.

3. Allow a smaller head or hand follow-through a frame or two after the torso settles.

WHY · Preparation makes the action legible; overshoot and settle suggest effort being absorbed.

CHECK / SUMMARY · The performance has a clear main action with a smaller recovery, rather than repeated bouncing.

FIX · If the settle distracts, reduce its amplitude and number of changes. Check the final pointing direction again.

GLOSSARY · anticipation: A preparation that helps viewers read the coming action.; overshoot: Passing slightly beyond the final position before returning.; settle: A decreasing recovery into rest.

DIFFERENT-OBJECT EXAMPLE · A person reaching for a shelf may shift weight first, reach beyond the target slightly and relax into the hold.

STRETCH · Make a cautious and an excited version using the same final pointing pose.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

07 · Diagnose the straight-line hand

SELECT: A fresh two-key comparison duplicate; preserve your designed version.

TOOLS / ROUTE: Delete selected transition keys only in the test copy.

1. In the comparison copy, keep only the thought and point endpoint keys for the transition. Remove the manual breakdown AND its manual in-betweens, then create classic spans with one matching Graphic instance on each animated part layer. Play this default interpolation and observe whether the hand cuts through the face or loses readability. If that particular pose pair does not cause an intersection, record the actual path/silhouette difference instead of inventing a fault.

2. Restore or redesign the breakdown and compare the wrist path with Onion Skin.

3. Explain the repair in terms of route and silhouette, not simply the number of frames.

WHY · The repair tests whether you understand what a breakdown contributes.

CHECK / SUMMARY · The repaired route has a visible clearance and retains the character's joint proportions.

FIX · If the route is still unclear, plot five wrist positions on a guide layer and redesign the curve before making more keys.

GLOSSARY · path of action: The route followed by a chosen part through the movement.

DIFFERENT-OBJECT EXAMPLE · A bouncing ball through a wall needs a different route, not just more in-between positions.

STRETCH · Make the repair with fewer transition poses while keeping the motion readable.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

08 · Direct the head-led reaction

SELECT: Your independent Think_Point variant.

TOOLS / ROUTE: New action duplicate; frame keys; full movie preview.

1. Add the changed-condition head turn before the pointing hand. Preserve the Caveman's eventual attention to the third stone.

2. Create keys, one labelled breakdown and at least two deliberate in-betweens. State which body part leads and which follows.

3. Save and reopen. Keep a pose strip, a timing note and your brief defence. Place the selected version in Discovery. Use a one-shot Graphic with First 1 and enough parent exposure; stop on a deliberate final pose instead of letting the phrase loop.

WHY · A new attention cue tests acting and sequence design rather than copying frame numbers.

CHECK / SUMMARY · The noise reaction is readable but does not replace the story's goal or create an unexplained extra event.

FIX · If the head and arm move as one stiff block, offset their starts and preserve a clear pause for recognition.

GLOSSARY · lead and follow: A deliberate order in which body parts start or finish movement.

DIFFERENT-OBJECT EXAMPLE · Independent task: use the same pose and timing criteria, without a supplied solution.

STRETCH · Use a subtle chest turn to support the head-led reaction.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

Keep a small, useful evidence record

At frame ___, I chose ___ so the viewer could ___.
The problem was caused by ___. I changed ___ and checked ___.
Under the changed condition, I revised ___ because ___.

One repair

One independent decision and its trade-off

Delivery check

The selected file reopens. The result plays as intended. Joints/contact and audio were checked where relevant. The changed condition is solved. Required production files and a brief defence are retained.

Method references and attribution

A1 · Adobe Animate: symbols and instances · Adobe reference

A2 · Adobe Animate: timeline layers · Adobe reference

A3 · Adobe Animate: shape tweening · Adobe reference

A4 · Adobe Animate: parenting view · Adobe reference

A5 · Adobe Animate: sound · Adobe reference

A6 · Adobe Animate: camera · Adobe reference

Original teacher-supplied Caveman PDF/PNG assets are preserved. New matching Caveman parts/backgrounds are AI-generated and identified in the asset register. Teaching diagrams are authored illustrations; genuine native captures are labelled separately. Supplied commercial course videos/FLAs are reference material and are not redistributed.

Publication/classroom status: see the teacher companion. This file does not certify the independent cold run, a natural voice performance, or live assessment/progress connections.

WORKBOOK 04 / 08 · MULTIMEDIA 12

Walk to the edge with weight

Contact, down, passing, up, cycles and planted feet

24-FRAME WALK · EIGHT DESIGNED POSES



1 · contact | 4 · down | 7 · passing | 10 · up | 13 · contact | 16 · down | 19 · passing | 22 · up

Loop frames 1–24. Frame 25 matches frame 1 only for construction.

Eight designed poses for a 24-frame cycle. The optional construction endpoint at 25 is not an extra loop hold.

Build: Create a readable walk cycle and stage it without obvious foot sliding.

Keep: Walk_24 Graphic + staged approach + one diagnosed foot-slip repair

Shared story: A Caveman wants a third counting stone. A stream stands in his way. He studies it, commits to a jump, retrieves the stone and returns to count his collection with enormous pride.

Document: ActionScript 3.0 · 1280 × 720 · 24 fps · three main Scenes · Graphic symbols for scrubbable character actions

Two provisional 70-minute sessions. Use the HTML guide for read-aloud, switchable supports, recovery files and narrated explanation. This print version includes the essential support prompts.

Continue the same film

Create the project in Workbook 1. For every later workbook, open the selected previous production FLA and Save As a new numbered checkpoint, preserving all three Scenes and the complete Library. Keep repair/director copies separate; return to the chosen production version.

Understand before you animate

A walk alternates support. The body moves over a planted foot while the other leg swings forward.

The eight design poses are contact, down, passing and up, then the opposite leg repeats them. In a 24-frame cycle use poses at 1, 4, 7, 10, 13, 16, 19 and 22. A matching pose at 25 can help construct the seam, but the repeating export/loop contains frames 1–24 so the endpoint is not held twice.

Today’s checklist

1. Predict the result. 2. Build four stages. 3. Check in playback. 4. Refine, diagnose and direct. 5. Save and reopen.

Independent director condition

After retrieving the stone, the Caveman takes a careful departure walk along the far bank before the return-to-cave cut. Adjust stride, balance and hand contact; do not put a third-stone carry before he has acquired it.

What you will defend

Point to the supporting foot and show how its world position stays consistent during stance.

Support choices

Use the step summaries, glossary, numbered checklist, different-object examples, explanation frames and optional stretch. Open the companion HTML for browser read-aloud and the narrated six-slide explanation. Supports change access to the task; the success criteria remain the same.

Your planning notes

01 · Mark support before moving parts

SELECT: A new Walk_24 Graphic; guide layer above the character.

TOOLS / ROUTE: Timeline keys; guide line; Onion Skin.

1. Draw a ground guide and mark hip height. Add a contact reference at frame 1 and opposite contact at frame 13.

2. Name the supporting leg at each key. Put the feet on the same ground plane and make the forward/back separation readable.

3. Use near/far identity consistently. The near leg does not become the far layer when it travels behind the other leg.

WHY · Support and identity must stay coherent through the cycle.

CHECK / SUMMARY · Both contact poses have clear foot placement and opposite leg arrangements.

FIX · If legs switch visual identity, restore their original layer names and use depth ordering consciously.

GLOSSARY · contact pose: A key moment when a foot meets the ground.; stance: The phase in which a foot supports weight on the ground.

DIFFERENT-OBJECT EXAMPLE · A table does not change which leg it is when viewed from another angle; limb identity stays consistent too.

STRETCH · Record a short observation of your own walk without collecting anyone else's image.

02 · Build down, passing and up

SELECT: Walk_24 frames 4, 7 and 10; then 16, 19 and 22.

TOOLS / ROUTE: Insert Keyframe on all part layers; Free Transform.

1. At frame 4 lower the hips slightly and bend the supporting knee. At 7 bring the swing foot past the support leg with toe clearance. At 10 raise the hips slightly as the support heel lifts.

2. At frames 16, 19 and 22 repeat these jobs with opposite legs. Keep head movement smaller than hip movement.

3. Counter-swing the arms: the arm opposite the forward leg comes forward. Avoid identical rotations on both sides.

WHY · The rise and fall comes from support mechanics; it is not a random vertical bob.

CHECK / SUMMARY · The swing foot clears the ground and the body height changes are restrained and rhythmic.

FIX · If knees pop backward, inspect the joint bend direction and add a passing breakdown with a clear knee shape.

GLOSSARY · passing pose: The swing leg passes the supporting leg.; up pose: The higher body position as support approaches push-off.

DIFFERENT-OBJECT EXAMPLE · A two-legged toy looks weightless if its torso never responds to support changes.

STRETCH · Make the proud Caveman’s chest slightly lead while keeping the pelvis balanced.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

03 · Fill the cycle and inspect the seam

SELECT: The gaps between the eight design poses.

TOOLS / ROUTE: Insert Keyframe; Onion Skin; loop playback.

1. Add the required in-between poses or short classic spans between the designed keys. Check feet and knees on every frame where interpolation is used.

2. Create a matching frame 25 temporarily to compare with frame 1. Compare pose and movement direction across 22→24→1.

3. Keep a separate 25-frame construction copy. If a classic span ends at 25, bake that span to frame-by-frame through 24 before removing the construction endpoint, or manually key 23 and 24. The final loop contains frames 1–24 with no duplicate endpoint hold.

WHY · A cycle must join in both pose and rhythm; matching only the first and last picture is not enough.

CHECK / SUMMARY · No pause or snap occurs at the loop boundary.

FIX · If there is a hitch, remove a duplicate contact hold or repair spacing across the boundary. Do not hide the seam under a camera cut before diagnosing it.

GLOSSARY · cycle: A repeatable motion sequence.; loop seam: The transition from the final frame back to the first.

DIFFERENT-OBJECT EXAMPLE · A rotating wheel can have identical endpoint pictures but still pause if both are held in the loop.

STRETCH · Test two consecutive cycles with the ground guide visible.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

04 · Stage the approach in Crossing

SELECT: Main Scene Crossing; Character layer.

TOOLS / ROUTE: Library → drag Walk_24 instance; Graphic → Loop; main timeline keys.

1. Place a Walk_24 Graphic on the near bank and set Loop. Give the main Scene enough frames for three cycles: 72 frames at 24 fps.

2. Move the action container from a starting point to the near-bank stopping point using keys at 1 and 73 as construction endpoints. Right-click the intervening span and Create Classic Tween, then inspect the travel and Graphic playback. Bake through frame 72 before replacing the construction endpoint at 73 with Stop, so the final tween does not lose its endpoint.

3. Keep overall scale constant during the walk. Do not translate a flattened still and call it a walk cycle.

WHY · The inner action supplies articulation; the outer motion supplies travel across the scene.

CHECK / SUMMARY · Leg motion and overall travel work together, and the Caveman reaches the bank rather than crossing the gap during the walk.

FIX · If the inner action freezes, check Graphic Loop and parent timeline length. If it races, inspect the fps and duplicated endpoint.

GLOSSARY · nested animation: An action timeline placed inside another timeline.; screen translation: Movement of the action container across the Stage.

DIFFERENT-OBJECT EXAMPLE · A bicycle wheel rotates inside a bicycle that also travels across the screen.

STRETCH · Compare a slow cycle with short travel against a fast cycle with long travel.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

05 · Match stride and travel to reduce sliding

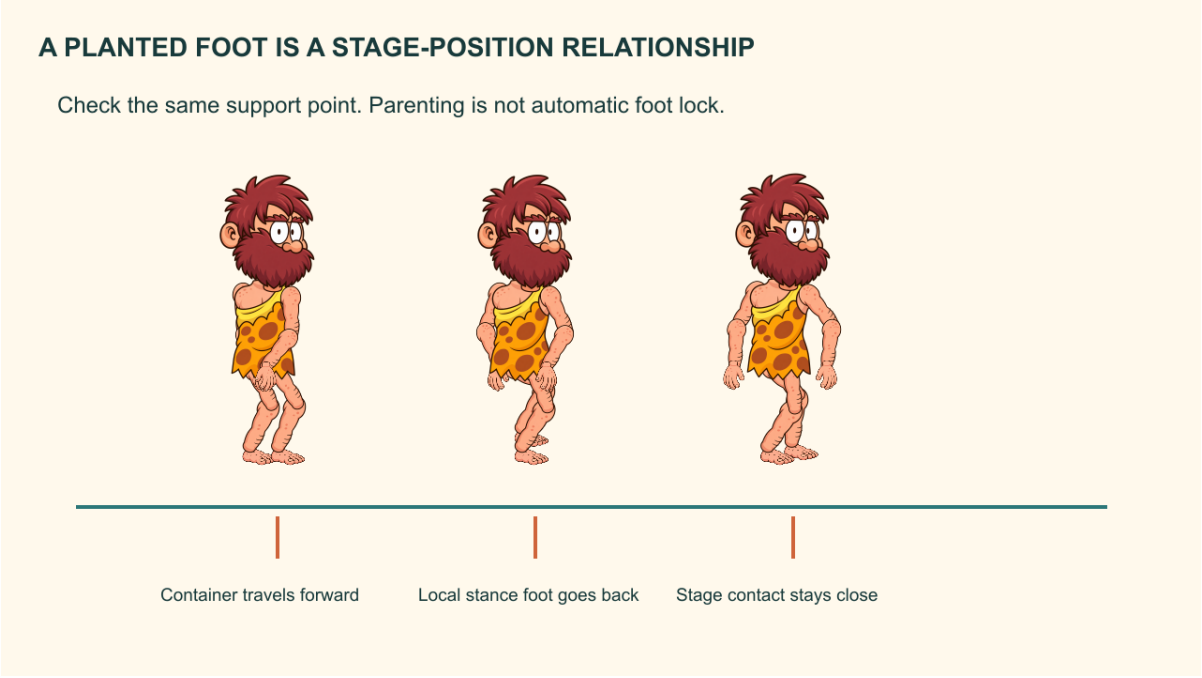
SELECT: A visible ankle/toe point during the stance half-cycle.

TOOLS / ROUTE: Onion Skin; frame-by-frame preview; guide marks.

1. Mark the supporting toe on the Stage at the start of stance. Advance several frames and compare its world position, including the outer container movement.

2. For the worked 24-frame cycle, at an unrotated 100% outer scale, a 12-frame stance with about 43 pixels of outer travel needs about 43 pixels of backward local foot travel to keep that foot planted. These are adjustable starting values, not universal anatomy. At 50% outer scale, about 86 local pixels produce 43 Stage pixels; judge the final Stage contact directly.

3. Adjust the local foot path, outer travel or stride duration together. Keep the foot at ground height through stance; lift it during swing.



Local foot movement and outer travel must agree during stance. Schematic comparison; assess actual Stage contact in playback.

WHY · A planted foot is a relationship between local leg motion and overall travel. Parenting alone does not lock it.

CHECK / SUMMARY · The support toe stays close to its guide mark until lift-off while the hips pass over it.

FIX · If it drifts forward, reduce outer speed or increase backward local foot motion. If it drifts backward, do the reverse. Recheck the other leg.

GLOSSARY · local position: A position within the action symbol.; world/Stage position: The final position after parent transforms are included.; foot slip: Unintended movement of a supposedly planted foot.

DIFFERENT-OBJECT EXAMPLE · A treadmill walker moves feet backward locally while staying in one place globally.

STRETCH · Measure slip in pixels over a stance and reduce it through one justified adjustment.

06 · Give the walk a motivated stop

SELECT: End of approach on the near bank.

TOOLS / ROUTE: Create a named 12-frame Graphic Stop from the neutral template; local keys and holds.

1. Create a Graphic named Stop with 12 local frames. Match local frame 1 to the final walk contact, shorten the final step, then plant the support foot before settling hips/chest over it. Use this named action at Crossing frames 73–84 in the worked assembly.

2. Let the arms finish their swing after the main body slows. Turn the head toward the far-bank stone.

3. Hold the assessment pose briefly before jump anticipation begins. The stop must not overlap an accidental looping walk frame.

WHY · Stopping needs a transfer of weight and a reason; an abrupt frozen cycle reads as an error.

CHECK / SUMMARY · The Caveman visibly judges the gap before preparing the jump.

FIX · If a foot pops, choose a compatible contact pose as the hand-off and repose the transition. Do not change the whole character's scale to reach the ground.

GLOSSARY · follow-through: A smaller continuation after the main action slows or stops.

DIFFERENT-OBJECT EXAMPLE · A runner's arms and clothing can finish moving after the feet stop.

STRETCH · Make a cautious stop and a confident stop with the same endpoint.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

07 · Repair one measured slip

SELECT: A duplicate approach with deliberately excessive outer travel.

TOOLS / ROUTE: Main timeline travel key; guide marks; before/after export.

1. Increase outer travel enough to create an obvious slide. Mark the support toe at three stance frames.

2. Restore plausible travel or revise the local stance path. Keep the cycle length unchanged for this comparison.

3. Retain three before/after toe positions and state which relationship you changed.

WHY · The repair shows whether you can diagnose the nested motion rather than just polish its appearance.

CHECK / SUMMARY · The repaired support point moves less while the body still reaches the intended stop.

FIX · If one foot improves but the other worsens, check the opposite half-cycle and consistent stride lengths.

GLOSSARY · constraint: A relationship that must be maintained, such as foot contact.

DIFFERENT-OBJECT EXAMPLE · A rolling wheel slips if its rotation and travel distance disagree.

STRETCH · Add a heel-to-toe roll while preserving the support-point logic.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

08 · Direct the carrying walk

SELECT: Your director action duplicate; stone attached by deliberate placement.

TOOLS / ROUTE: Pose keys; hand/prop layers; main scene travel.

1. After pickup, keep the stone held close while the Caveman walks carefully along the far bank. Reduce stride or hip bounce and revise balance rather than only changing playback speed.

2. Check hand-to-stone contact on every designed pose and stance-foot contact during travel.

3. Save and reopen. Explain one performance change and one contact repair, then use the carrying version only for the optional far-bank departure after pickup. Retain the empty-handed approach before the jump.

WHY · Carrying changes body mechanics and intention; it requires a new performance decision.

CHECK / SUMMARY · The prop stays held and the walk reads as careful under the same story goal.

FIX · If the prop floats, key it with the carrying hand or place it in the appropriate shared container and verify that relationship.

GLOSSARY · secondary action: A supporting action that enriches the main action without obscuring it.

DIFFERENT-OBJECT EXAMPLE · Independent task: use support, contact and silhouette criteria without a supplied carrying cycle.

STRETCH · Add a brief glance at the stone while preserving safe foot placement.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

Keep a small, useful evidence record

At frame ___, I chose ___ so the viewer could ___.
The problem was caused by ___. I changed ___ and checked ___.
Under the changed condition, I revised ___ because ___.

One repair

One independent decision and its trade-off

Delivery check

The selected file reopens. The result plays as intended. Joints/contact and audio were checked where relevant. The changed condition is solved. Required production files and a brief defence are retained.

Method references and attribution

A1 · Adobe Animate: symbols and instances · Adobe reference

A2 · Adobe Animate: timeline layers · Adobe reference

A3 · Adobe Animate: shape tweening · Adobe reference

A4 · Adobe Animate: parenting view · Adobe reference

A5 · Adobe Animate: sound · Adobe reference

A6 · Adobe Animate: camera · Adobe reference

Original teacher-supplied Caveman PDF/PNG assets are preserved. New matching Caveman parts/backgrounds are AI-generated and identified in the asset register. Teaching diagrams are authored illustrations; genuine native captures are labelled separately. Supplied commercial course videos/FLAs are reference material and are not redistributed.

Publication/classroom status: see the teacher companion. This file does not certify the independent cold run, a natural voice performance, or live assessment/progress connections.

WORKBOOK 05 / 08 · MULTIMEDIA 12

Jump the stream with a designed arc

Anticipation, take-off, breakdowns, landing, overshoot and settle

A JUMP CHANGES POSE AND SUPPORT



1 · ready | 9 · compress | 12 · take-off | 19 · apex | 28 · contact | 36 · recover

Stepped keys first → breakdowns → intentional spacing → recovery.

Jump phase studies: preparation, extension, flight, contact and recovery.

Build: Animate a full jump with readable weight and a controlled landing.

Keep: Jump_36 Graphic + four-to-seven pose strip + revised landing

Shared story: A Caveman wants a third counting stone. A stream stands in his way. He studies it, commits to a jump, retrieves the stone and returns to count his collection with enormous pride.

Document: ActionScript 3.0 · 1280 × 720 · 24 fps · three main Scenes · Graphic symbols for scrubbable character actions

Two provisional 70-minute sessions. Use the HTML guide for read-aloud, switchable supports, recovery files and narrated explanation. This print version includes the essential support prompts.

Continue the same film

Create the project in Workbook 1. For every later workbook, open the selected previous production FLA and Save As a new numbered checkpoint, preserving all three Scenes and the complete Library. Keep repair/director copies separate; return to the chosen production version.

Understand before you animate

The jump is a change in support: grounded preparation, flight, impact and recovery. The pose and spacing must explain each phase.

Use a provisional 36-frame action: anticipation at 1 and 9; take-off at 12; apex near 19; first landing near 28; settle by 36. These are authored teaching timings, not cues from the old 105-frame lip-sync test. The root travels along an arc while body parts change pose. Uniform easing over the whole jump does not model these phases.

Today’s checklist

1. Predict the result. 2. Build four stages. 3. Check in playback. 4. Refine, diagnose and direct. 5. Save and reopen.

Independent director condition

The stream is wider. Keep the character and goal, then revise anticipation, travel, flight timing and landing reach. Do not stretch the entire picture horizontally.

What you will defend

Show the take-off, apex and first-contact frames and explain one spacing decision.

Support choices

Use the step summaries, glossary, numbered checklist, different-object examples, explanation frames and optional stretch. Open the companion HTML for browser read-aloud and the narrated six-slide explanation. Supports change access to the task; the success criteria remain the same.

Your planning notes

01 · Plan contact, flight and landing

SELECT: Jump planning strip and a new Jump_36 Graphic.

TOOLS / ROUTE: Guide layer; timeline labels or notes; held pose keys.

1. Mark the near-bank take-off and far-bank landing points on the stream background. Draw a root/hip arc above them.

2. Block standing at 1, deepest anticipation at 9, extended take-off at 12, apex at 19, first landing at 28 and recovered stance at 36.

3. Keep the first version stepped. Check that the landing point is reachable and that the intended take-off direction is clear.

WHY · The action needs a destination and a sequence of support states before it needs smooth movement.

CHECK / SUMMARY · The body clears the stream and lands on the far bank with feet directed toward the ground.

FIX · If the jump appears vertical, increase forward travel and revise the torso lean and leg extension toward the destination.

GLOSSARY · root: A chosen overall body reference, often near the hips.; apex: The highest point of the flight path.

DIFFERENT-OBJECT EXAMPLE · A thrown ball needs a launch point, flight path and destination before its in-betweens are placed.

STRETCH · Design a low athletic jump and a high cautious jump as stepped strips.

02 · Build the anticipation as compression

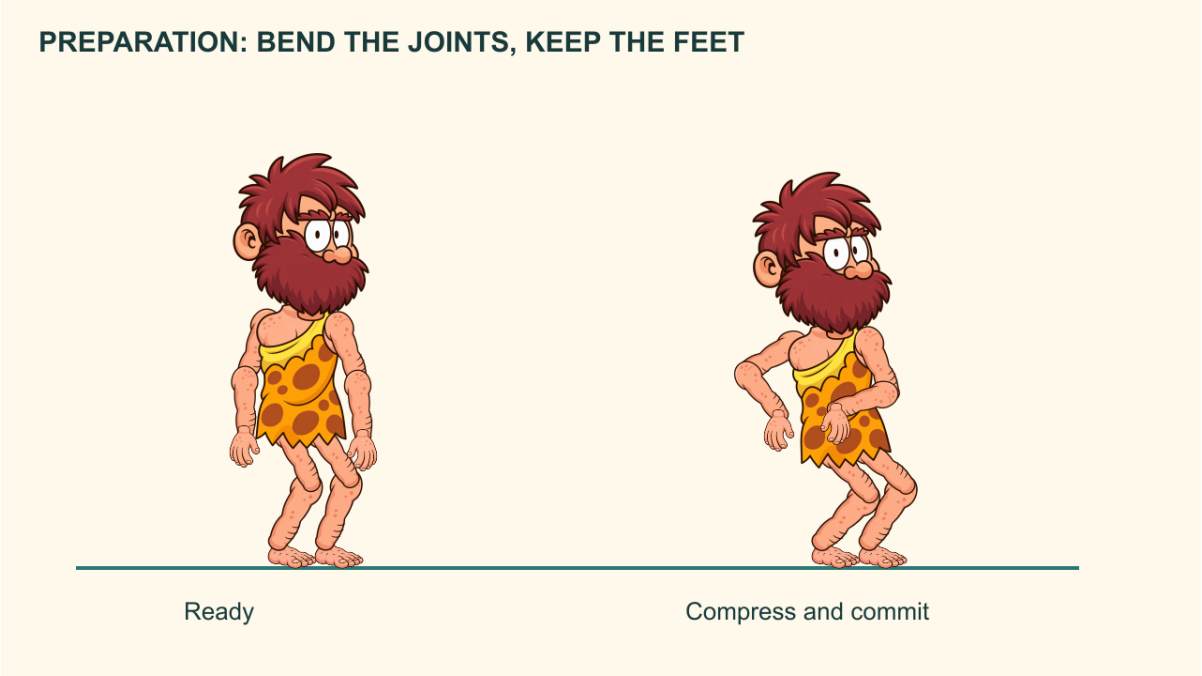
SELECT: Jump_36 frames 1 and 9; all articulated body layers.

TOOLS / ROUTE: Insert Keyframe; Free Transform; ground guide.

1. At frame 9 bend knees and hips, lean the torso toward the destination and move arms back. Keep both feet planted.

2. Lower the hips by posing the legs, not merely scaling the whole Caveman shorter. Keep the head looking where he will land.

3. Use tighter spacing near the deepest crouch if you want a brief commitment before release.



Preparation through bent joints, with feet retained on the ground plane.

WHY · Anticipation stores visual energy and makes the direction of the coming action readable.

CHECK / SUMMARY · The crouch has bent joints, grounded feet and a clear forward intention.

FIX · If the soles lift during the crouch, reposition the hips and knees while preserving foot positions. Reduce the crouch if the rig cannot maintain believable proportions.

GLOSSARY · compression: A pose in which the body folds to prepare or absorb force.; anticipation: The preparation before the main action.

DIFFERENT-OBJECT EXAMPLE · A spring compresses before release; a human jump prepares through joints and weight shift.

STRETCH · Compare a quick dip with a longer uncertain preparation.

03 · Design take-off, an in-flight breakdown and the apex key

SELECT: Jump_36 frames 12, 16 and 19.

TOOLS / ROUTE: Pose keys; Onion Skin; joint repositioning.

1. At take-off extend hips, knees and ankles toward the jump direction. Let the arms swing through to support the action.

2. At frame 16 design an in-flight breakdown between take-off and the apex key: gather the knees and choose the arm path while keeping clear silhouettes. At the frame 19 apex key, show the top-of-flight pose and its different support state.

3. Keep limb lengths consistent. The frame 16 breakdown decides the transition; the apex remains a key. Neither role is defined by being an automatic midpoint.

WHY · The flight pose tells the audience how the character gets across, while take-off explains the launch.

CHECK / SUMMARY · There is a visible change from compressed to extended to gathered, with no disconnected joints.

FIX · If the body looks like a stiff sticker, change the torso angle, knee bend and arm relationship across these phases.

GLOSSARY · extension: Opening the joints into a longer pose.; flight breakdown: A designed in-air pose defining how take-off connects to landing.

DIFFERENT-OBJECT EXAMPLE · A frog extends at launch and gathers its legs during flight rather than keeping one pose throughout.

STRETCH · Let the head lag slightly after take-off, then lead toward the landing.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

04 · Space the root along the arc

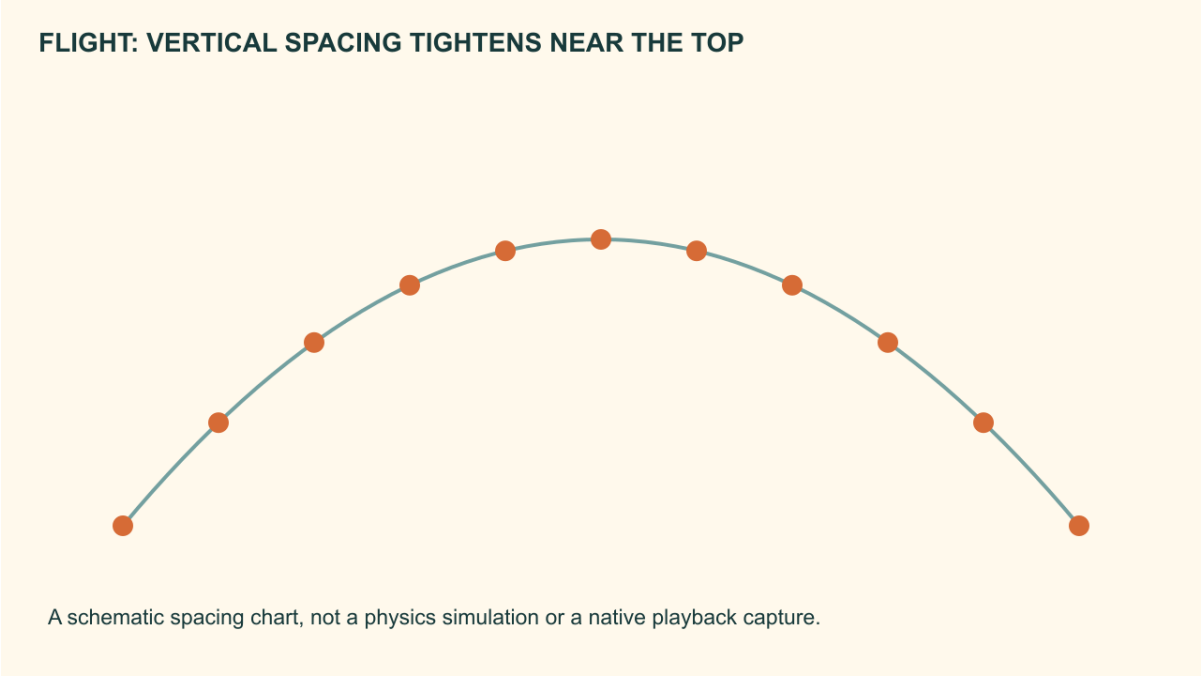
SELECT: Root/whole action placement through frames 12–28.

TOOLS / ROUTE: Inside Jump_36, select all fourteen part-layer keys together for the root offset; keep the outer Stage instance fixed. Guide arc; frame-by-frame preview.

1. Build the root travel inside Jump_36 by moving all fourteen part-layer poses together at each designed frame. Keep its outer Crossing instance fixed during playback. Place intermediate root positions along the planned arc. Use smaller vertical changes near the apex and larger vertical changes during faster ascent/descent.

2. Keep forward travel reasonably consistent through flight unless your deliberate style calls for otherwise. Do not apply a single strong ease-in/ease-out to the entire jump.

3. At each important root position, repose limbs as required. Inspect the trajectory in context with the stream banks.



Schematic trajectory study: vertical displacement per frame becomes smaller near the apex. Not a physics simulation.

WHY · Timing controls flight duration; spacing communicates changing speed. The body also changes shape while following the path.

CHECK / SUMMARY · The jump rises and falls smoothly, slows vertically near the top and does not hover accidentally.

FIX · If the path makes a sharp corner, move the intermediate root point onto a continuous curve. If the landing floats, diagnose a pause or slow descent: reduce an unintended hold, increase downward position spacing toward impact where appropriate, and correct the actual contact pose.

GLOSSARY · trajectory: The path of an object or body through space.; spacing chart: A plan showing successive positions along a path.

DIFFERENT-OBJECT EXAMPLE · A ball moves only a little vertically near its apex but much farther per frame lower in its arc.

STRETCH · Keep the same arc but change flight duration and describe the change in apparent weight.

05 · Absorb the landing, then settle

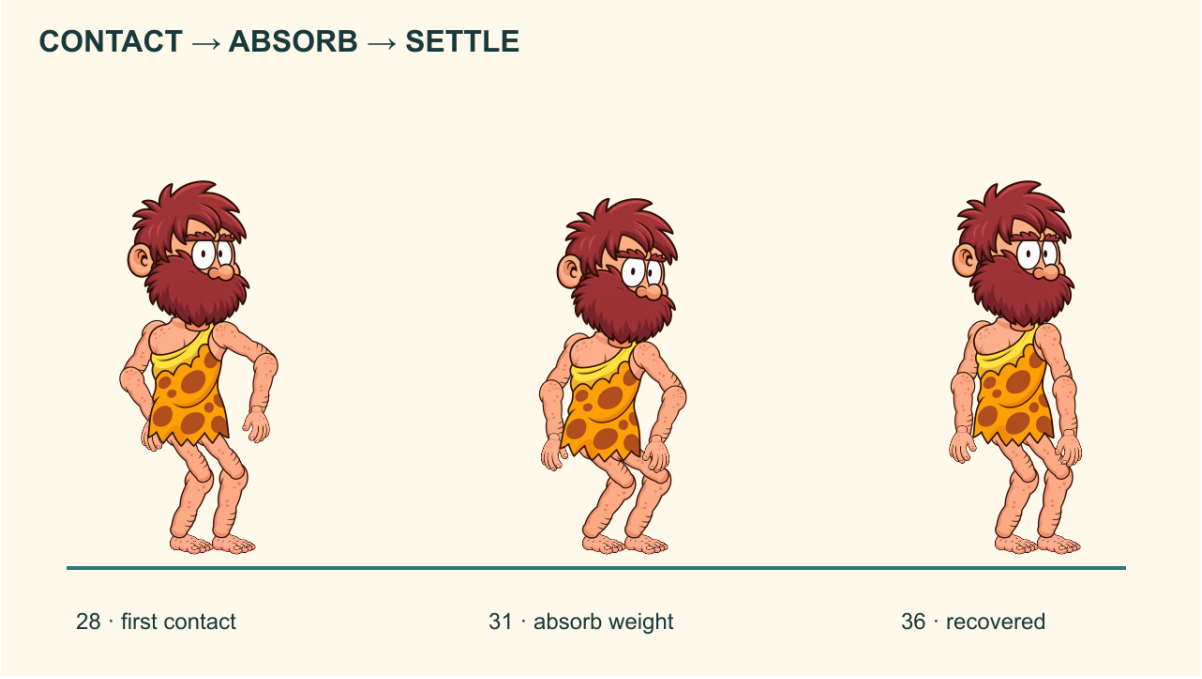
SELECT: Frames 28, 31 and 36; feet and hips visible.

TOOLS / ROUTE: Insert Keyframe; Free Transform; contact guide.

1. At first contact, place the leading foot on the far bank. At 31 bend knees and hips to absorb the landing; keep the contact foot fixed.

2. Let the torso and arms continue slightly past their final pose, then recover by 36 with a smaller movement.

3. Use restrained squash/stretch only as an optional accent. Preserve apparent body volume and facial readability; joint posing must still explain the landing.



First contact, absorption and decreasing recovery are separate pose jobs.

WHY · Impact requires a response. A straight-legged frozen landing removes weight and can look painful.

CHECK / SUMMARY · The contact foot stays grounded, the body compresses after contact and the recovery decreases in size.

FIX · If the Caveman bounces repeatedly, reduce the overshoot and number of recovery keys. If he sinks through the bank, restore contact before adjusting the torso.

GLOSSARY · squash and stretch: Controlled shape change that suggests force and flexibility while preserving apparent volume.; overshoot and settle: A small pass beyond rest followed by decreasing recovery.

DIFFERENT-OBJECT EXAMPLE · A soft ball compresses at impact; a heavy box may show its impact through a brief recoil and a strong stop instead.

STRETCH · Make a heavy landing without enlarging the character's head or changing its identity.

06 · Connect the jump to retrieving the stone

SELECT: A named 36-frame Pickup Graphic after the settled landing; stone contact separate from body parts.

TOOLS / ROUTE: New pickup pose keys; hand and prop contact.

1. Create a 36-frame Graphic named Pickup from the character template. Match local frame 1 to the settled jump pose. Bend through knees/hips and reach the hand to the stone; use local frame 25 as the worked grasp cue, then lift and recover by 36.

2. Keep the bank stone on the main Props layer until the grasp. At Pickup local 25 (Crossing 145 in the worked map), blank the bank copy and reveal a stone inside Pickup aligned to the hand. From then on animate that carried copy with the hand. Avoid displaying both copies.

3. Finish with a clear held-stone pose that supports the later cut back to the cave.

WHY · The jump serves the story goal. Retrieval proves the action achieved something.

CHECK / SUMMARY · The stone remains on the bank until contact and stays held afterward.

FIX · If the stone moves before the hand reaches it, move the prop key to the actual grasp frame. If it slides in the hand, inspect their relative positions.

GLOSSARY · contact frame: The precise frame at which two elements first meet.; continuity of props: Keeping an object's location and ownership understandable.

DIFFERENT-OBJECT EXAMPLE · A cup remains on a table until the fingers contact it, then follows the lift.

STRETCH · Add a brief effort pose before lifting a heavier-looking stone.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

07 · Repair a weightless jump

SELECT: A duplicate using one unchanged body pose along a straight line.

TOOLS / ROUTE: Compare playback; restore designed keys and arc.

1. Play the deliberately poor version. Identify two causes: an unchanged pose, missing anticipation, uniform spacing, no arc or an unabsorbed landing.

2. Repair those two causes without adding decorative effects. Keep a before/after pose strip.

3. Explain which change improves intention and which improves weight.

WHY · Principles are tools for diagnosing an action, not labels to add after it is finished.

CHECK / SUMMARY · The repair shows phase changes and a grounded landing; the explanation cites visible frames.

FIX · If the result is only smoother, return to the held keys and strengthen the support changes.

GLOSSARY · weight: The impression of mass created by timing, pose, support and response to force.

DIFFERENT-OBJECT EXAMPLE · A bowling ball and a balloon can follow similar routes but require very different spacing and recovery.

STRETCH · Reduce the number of keys while keeping all four support phases readable.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

08 · Direct the wider-stream version

SELECT: Your independent Jump_Wide Graphic and matching background staging.

TOOLS / ROUTE: Duplicate action; revise keys, travel and timing.

1. Widen the gap by an amount your character can plausibly clear. Revise the preparation, flight route and landing pose together.

2. Keep a labelled anticipation key, take-off key, flight breakdown, first-contact key and settle. Add intentional in-betweens.

3. Save and reopen. Show the two versions and defend one timing or spacing decision. Choose the version that belongs in the film.

WHY · A new distance requires a new movement solution and an explanation of the trade-off.

CHECK / SUMMARY · The character clears the changed obstacle and retains believable support and proportion.

FIX · If you need impossible reach, revise the staging or introduce a deliberate stylized exaggeration and justify it; do not silently stretch anatomy.

GLOSSARY · exaggeration: A purposeful amplification that makes an idea clearer while preserving its logic.

DIFFERENT-OBJECT EXAMPLE · Independent task: use the same support and arc criteria; no wider-gap pose solution is supplied.

STRETCH · Make the wider jump feel risky without adding extra dialogue.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

Keep a small, useful evidence record

At frame ___, I chose ___ so the viewer could ___.
The problem was caused by ___. I changed ___ and checked ___.
Under the changed condition, I revised ___ because ___.

One repair

One independent decision and its trade-off

Delivery check

The selected file reopens. The result plays as intended. Joints/contact and audio were checked where relevant. The changed condition is solved. Required production files and a brief defence are retained.

Method references and attribution

A1 · Adobe Animate: symbols and instances · Adobe reference

A2 · Adobe Animate: timeline layers · Adobe reference

A3 · Adobe Animate: shape tweening · Adobe reference

A4 · Adobe Animate: parenting view · Adobe reference

A5 · Adobe Animate: sound · Adobe reference

A6 · Adobe Animate: camera · Adobe reference

Original teacher-supplied Caveman PDF/PNG assets are preserved. New matching Caveman parts/backgrounds are AI-generated and identified in the asset register. Teaching diagrams are authored illustrations; genuine native captures are labelled separately. Supplied commercial course videos/FLAs are reference material and are not redistributed.

Publication/classroom status: see the teacher companion. This file does not certify the independent cold run, a natural voice performance, or live assessment/progress connections.

WORKBOOK 06 / 08 · MULTIMEDIA 12

Give the Caveman a voice

Recording, Stream sound, four mouth poses and true hand swaps

MANUAL FOUR-POSE GRAPHIC BANK



1 · closed | 2 · round | 3 · wide | 4 · teeth

Align visible lip centres. Single Frame holds the selected pose.

The original four mouth drawings, aligned by visible lip centre inside Mouth_Poses.

Build: Synchronize mouth and gesture changes to the actual recording and preserve the original Caveman identity.

Keep: Me_clever Scene + original voice recording + measured cue sheet + lip-sync revision

Shared story: A Caveman wants a third counting stone. A stream stands in his way. He studies it, commits to a jump, retrieves the stone and returns to count his collection with enormous pride.

Document: ActionScript 3.0 · 1280 × 720 · 24 fps · three main Scenes · Graphic symbols for scrubbable character actions

Two provisional 70-minute sessions. Use the HTML guide for read-aloud, switchable supports, recovery files and narrated explanation. This print version includes the essential support prompts.

Continue the same film

Create the project in Workbook 1. For every later workbook, open the selected previous production FLA and Save As a new numbered checkpoint, preserving all three Scenes and the complete Library. Keep repair/director copies separate; return to the chosen production version.

Understand before you animate

Lip sync follows heard sound shapes and phrase emphasis. Gestures support meaning; they do not have to change on every mouth cue.

The original four-mouth method is manual Graphic Single Frame selection: closed, round, wide and teeth. It is a simplified stylized bank, not a full phoneme set. The hand images are three separate Graphic definitions changed with Swap Symbol. The previous 105-frame file held one round mouth; it was a mechanism test, not a completed lip-sync performance. A synthetic test WAV is labelled technical reference only.

Today’s checklist

1. Predict the result. 2. Build four stages. 3. Check in playback. 4. Refine, diagnose and direct. 5. Save and reopen.

Independent director condition

Record a more doubtful or more triumphant reading of the same line. Retain the words, remeasure cues, and redesign the gesture and timing.

What you will defend

Show one audible cue, its timeline frame and the selected mouth pose; then explain one deliberate hold.

Support choices

Use the step summaries, glossary, numbered checklist, different-object examples, explanation frames and optional stretch. Open the companion HTML for browser read-aloud and the narrated six-slide explanation. Supports change access to the task; the success criteria remain the same.

Your planning notes

01 · Record the actual performance first

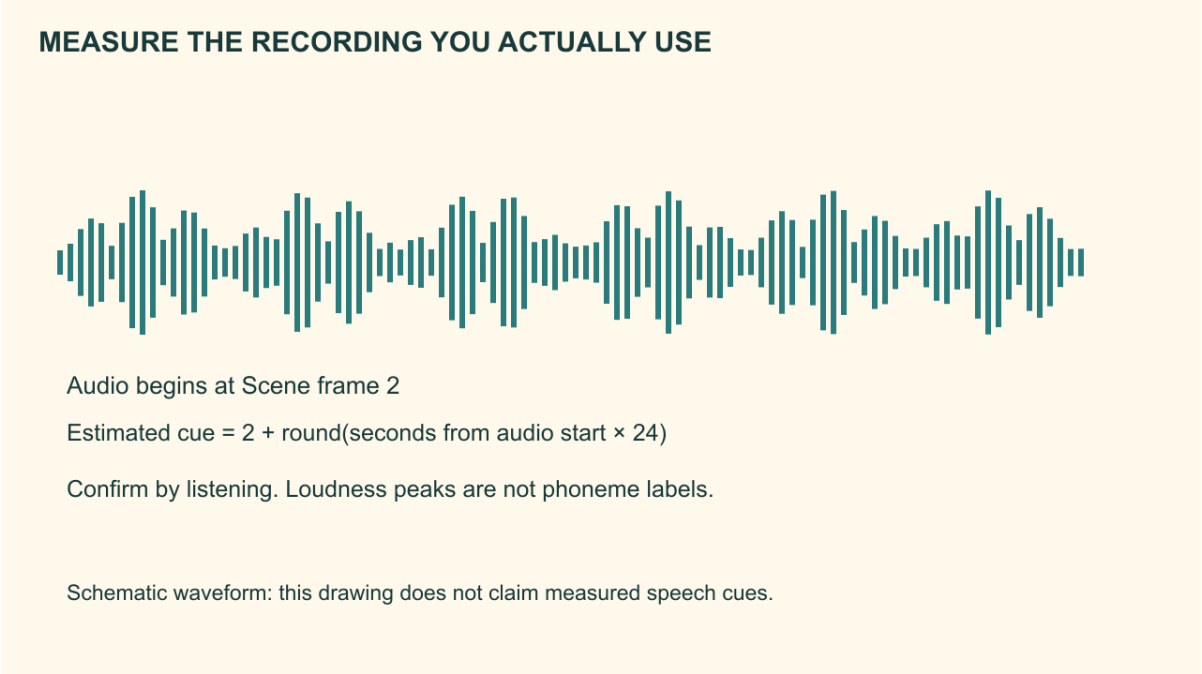
SELECT: A recording application approved in your classroom; then Animate's Library.

TOOLS / ROUTE: Record → trim → export PCM WAV; File → Import → Import to Library.

1. Record "Me count! One… two… three! Me clever!" with a clear pause between numbers, or use the teacher's actual supplied take. Listen to the whole recording before using it.

2. Trim only unwanted silence, keeping a little lead-in. Export a standard PCM WAV and give it a meaningful name. Preserve the original take.

3. Import that recording. The supplied synthetic test sound can test the mechanism, but it is not a natural performance approved for the film.



Cue-sheet method diagram. The waveform is schematic; students measure and listen to their actual take.

WHY · The recorded performance determines cue timing and emotional rhythm.

CHECK / SUMMARY · The file plays clearly with the intended words and no clipped start or end.

FIX · If speech is distorted, rerecord at a lower input level. If the import fails, convert the audio to a supported PCM WAV with the classroom's approved audio tool.

GLOSSARY · clipping: Distortion caused by an audio level exceeding the recording limit.; PCM WAV: An uncompressed audio file format useful for an editing workflow.

DIFFERENT-OBJECT EXAMPLE · A narrator's pause before a reveal affects the edit; the edit should respect the actual take.

STRETCH · Record two contrasting performances and choose based on the story ending.

02 · Place Stream sound and measure the cues

SELECT: Me_clever main Scene; Audio layer frame 2.

TOOLS / ROUTE: Insert Keyframe at 2; Properties → Frame → Sound → Sync: Stream.

1. Create a dedicated Audio layer and place the recording on frame 2. Set Sync to Stream in Frame properties. Measure the audio end, add a short ending hold, and extend the Audio layer AND every visible layer to that same final frame. An early-ending layer must not make a body part disappear.

2. Scrub and listen. On a guide Notes layer record the frame where each phrase/number begins, the strongest vowel and the close or silence.

3. For a timestamp t seconds from the start of the audio, a first estimate is frame 2 + round(t × 24). Confirm by listening; waveform peaks alone do not identify phonemes.

WHY · Stream sound supports timeline synchronization. Real cues must come from this take, not the old 1/25/49 control test.

CHECK / SUMMARY · All words play in the intended order, the clip is not cut off and the cue sheet names actual observed frames.

FIX · If the start is silent, confirm the sound key is at 2 and that enough frames exist. If scrubbing is unreliable, test playback and inspect the selected frame's sound settings.

GLOSSARY · Stream sound: Sound synchronized to timeline playback.; cue sheet: A record linking heard events to timeline positions.

DIFFERENT-OBJECT EXAMPLE · A clap can be estimated from a waveform spike, then confirmed by listening at the marked frame.

STRETCH · Mark the phrase stress separately from the mouth changes.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

03 · Build and align the four-pose mouth bank

SELECT: A new Graphic named Mouth_Poses.

TOOLS / ROUTE: Library → New Symbol → Graphic; internal frames 1–4.

1. Inside Mouth_Poses, put closed at internal frame 1, round at 2, wide at 3 and teeth at 4, with one keyframe per pose.

2. Align all mouths to the same visible lip centre. Source PNG canvases have different transparent margins, so matching their raw X/Y values is not enough.

3. Place one Mouth_Poses instance on a separate Mouth layer over the original caveman_base beard's mouth area in Me_clever. Keep the body on its own locked layer.

WHY · A shared mouth centre prevents lip sync from becoming positional jitter.

CHECK / SUMMARY · Cycling through all four internal frames changes the mouth shape while its visible centre stays in place.

FIX · If the mouth jumps, correct alignment inside the bank. Do not compensate by moving every main-timeline mouth instance separately.

GLOSSARY · mouth bank: A symbol containing a set of reusable mouth drawings.; visible alignment: Matching the artwork itself rather than transparent canvas edges.

DIFFERENT-OBJECT EXAMPLE · Four eye drawings should align by the pupil/eyelid position, not by image canvas corners.

STRETCH · Add a neutral hold using the existing closed pose before creating any extra mouth artwork.

04 · Set manual Single Frame lip-sync keys

SELECT: The placed Mouth_Poses instance on the main Scene timeline.

TOOLS / ROUTE: Properties → Object → Looping: Single Frame; Frame picker or First.

1. First initialize the main Mouth layer at frame 1: select Mouth_Poses, choose Single Frame and First 1 (closed). This prevents looping during the lead-in. At the first speech cue insert a new Mouth keyframe and choose its required pose with Frame picker or First.

2. Add keys at meaningful sound-shape changes. Use round for rounded vowels, wide for broad/open vowels, teeth where the stylized drawing fits and closed for silence or closed lips.

3. Replay in short phrases with sound. Prefer clear selected shapes and holds over a new random mouth on every frame. The four-pose bank approximates speech; it cannot draw every phoneme exactly.

WHY · Single Frame holds the selected internal drawing until the next main-timeline change.

CHECK / SUMMARY · The selected object is Mouth_Poses, not Hand_1. Different main-timeline keys show different First values and corresponding visible mouths.

FIX · If the mouth cycles automatically, choose Single Frame. If all cues show the same pose, check that you inserted main-timeline keys and changed the selected instance at each key.

GLOSSARY · Single Frame: A Graphic playback setting that holds one chosen internal frame.; phoneme: A speech sound; a drawing usually represents a family of related mouth shapes.

DIFFERENT-OBJECT EXAMPLE · A simplified talking fox can hold a wide mouth through a sustained vowel rather than cycling all shapes.

STRETCH · Lead or delay a mouth cue by one frame, compare and justify the clearer result.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

05 · Swap three separate counting hands

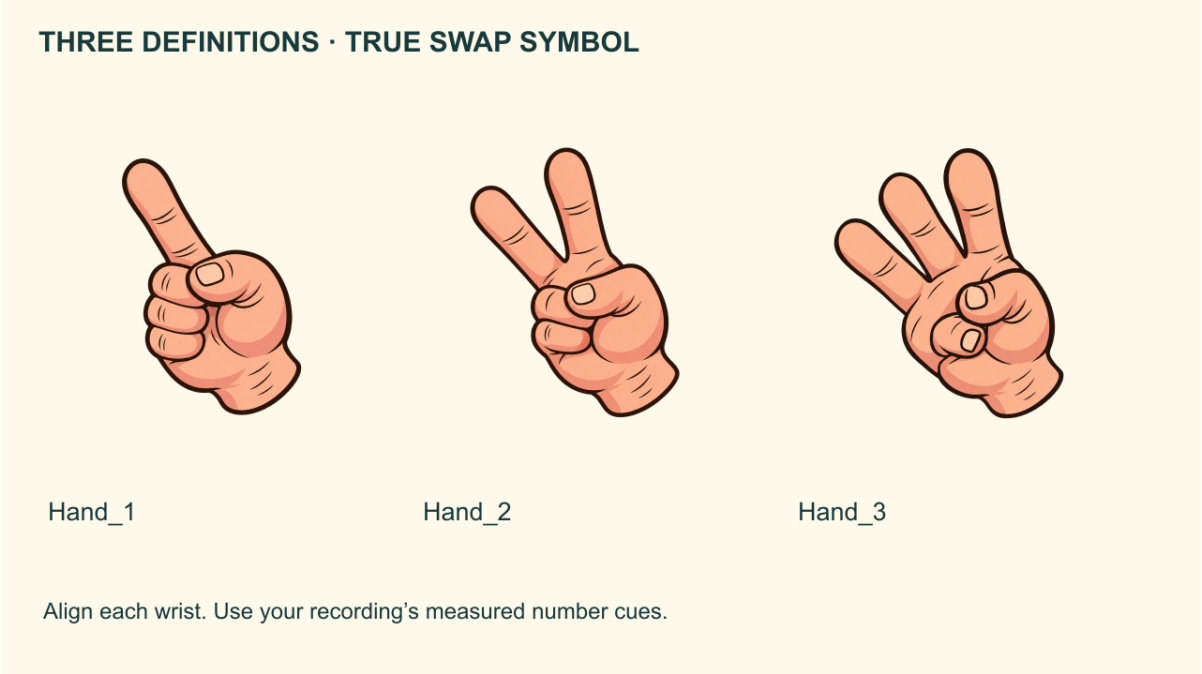
SELECT: The Hand layer in Me_clever; three named hand Graphic definitions.

TOOLS / ROUTE: Modify → Convert to Symbol; Properties → Swap Symbol.

1. Create separate Hand_1, Hand_2 and Hand_3 Graphic definitions from the supplied one-, two- and three-finger images. Align each visible wrist at a common registration location.

2. Place Hand_1 for “one.” Insert a new Hand key at your measured “two” gesture cue, select the instance and Swap Symbol to Hand_2. Repeat for Hand_3 at “three.”

3. Check and correct scale/position after each swap so the wrist remains attached. Keep this as true definition swapping, not a three-frame hand bank disguised as Swap.



Original one-, two- and three-finger artwork belongs to three separate Graphic definitions for true Swap Symbol.

WHY · Gestures communicate counting and can prepare slightly before the emphasized word.

CHECK / SUMMARY · Properties identifies Hand_1, then Hand_2, then Hand_3 at the measured cue keys; the wrist stays fixed.

FIX · If the hand jumps, align the wrist inside each definition and recheck the placed instance transforms. Do not use arbitrary 1/25/49 timings for a different recording.

GLOSSARY · Swap Symbol: Replacing an instance’s referenced Library definition.; gesture cue: The chosen moment at which a meaningful hand action occurs.

DIFFERENT-OBJECT EXAMPLE · A traffic light can swap distinct red/green symbols while retaining its mounting point.

STRETCH · Let the hand prepare before the number, then settle during the word.

06 · Add acting without obscuring the count

SELECT: Body/head action and hand transitions in the counting close-up.

TOOLS / ROUTE: Pose keys; held poses; phrase playback.

1. On “Me count,” establish focus on the stones. During the count keep each finger display readable long enough to understand.

2. For the original flattened seated setup, select the body, mouth and hand layers’ matching frames and Copy Frames into a new Graphic named Count_Performance, preserving the separate inner layers and all cue keys. Verify the copied keys and clear the original visible body, mouth and hand exposures on the main Scene so no duplicate remains beneath the container. Keep Audio on the main Scene. Place Count_Performance from main frame 1 with Play Once/First 1. On “Me clever,” use a restrained whole-container tilt or upward accent and settle, so the body, mouth and hand remain attached. On every added outer key at main frame N, set First to N because this container starts at main frame 1; this preserves the inner cue timing instead of restarting it. Do not claim an independently rotating head on this baked base.

3. Blink only if you have a genuine separate eye/eyelid setup. Do not paste another pair of eyes over baked eyes and call it a finished blink rig.

WHY · Secondary acting should reinforce the words and the result of the story.

CHECK / SUMMARY · The viewer can count all three fingers/stone beats and read pride at the end.

FIX · If everything moves on every syllable, remove unnecessary keys. Preserve one main emphasis per phrase.

GLOSSARY · phrase emphasis: The strongest intended meaning or stress within a spoken phrase.; moving hold: A restrained living pose that preserves the main idea.

DIFFERENT-OBJECT EXAMPLE · A speaker can hold a pointing gesture through an important noun while a small head movement supplies life.

STRETCH · Design an intentionally awkward proud pause after the last word.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

07 · Repair a deliberately mistimed phrase

SELECT: A duplicate lip-sync section, three to five frames out of time.

TOOLS / ROUTE: Select Mouth keys; shift keys; compare with audio.

1. Offset one phrase’s mouth keys enough to make the mismatch visible. Leave the audio fixed.

2. Replay, identify whether the mouth leads or lags, then restore alignment using the actual heard cue.

3. Retain one before/after frame pair with the cue-sheet entry and explain why still images alone cannot prove synchronization.

WHY · The repair tests listening and timing, not just control settings.

CHECK / SUMMARY · Playback of the corrected phrase agrees with the recording and the cue sheet.

FIX · If correcting one word breaks the next, select only the intended keys and preserve their internal rhythm.

GLOSSARY · lead/lag: An event occurring before/after the intended synchronized event.

DIFFERENT-OBJECT EXAMPLE · A door slam picture and sound must be checked in playback even when their timeline keys look close.

STRETCH · Compare a held vowel with excessive mouth chatter and explain which reads better.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

08 · Direct a new vocal reading

SELECT: Your director recording and duplicate counting Scene.

TOOLS / ROUTE: Import take; Stream sound; remeasure cues; mouth and hand keys.

1. Record the changed emotional reading of the same words. Replace the audio in the duplicate and measure its cues again.

2. Revise mouth selections, hand swaps and the final acting accent to match the new take. Do not slide the old key pattern blindly.

3. Save, close and reopen, then listen to the entire Scene. Keep the recording, cue sheet and a playable preview together.

WHY · A new performance requires new synchronization and acting decisions.

CHECK / SUMMARY · The selected performance is intelligible, synchronized and consistent with the film's ending.

FIX · If the new take changes length, extend or trim all relevant Scene layers and the ending hold deliberately.

GLOSSARY · performance timing: The rhythm supplied by a particular delivery of the line.

DIFFERENT-OBJECT EXAMPLE · Independent task: no cue sheet is supplied for a recording that does not yet exist.

STRETCH · Explain why one held mouth or delayed hand gesture supports the emotional reading.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

Keep a small, useful evidence record

At frame ___, I chose ___ so the viewer could ___.
The problem was caused by ___. I changed ___ and checked ___.
Under the changed condition, I revised ___ because ___.

One repair

One independent decision and its trade-off

Delivery check

The selected file reopens. The result plays as intended. Joints/contact and audio were checked where relevant. The changed condition is solved. Required production files and a brief defence are retained.

Method references and attribution

A1 · Adobe Animate: symbols and instances · Adobe reference

A2 · Adobe Animate: timeline layers · Adobe reference

A3 · Adobe Animate: shape tweening · Adobe reference

A4 · Adobe Animate: parenting view · Adobe reference

A5 · Adobe Animate: sound · Adobe reference

A6 · Adobe Animate: camera · Adobe reference

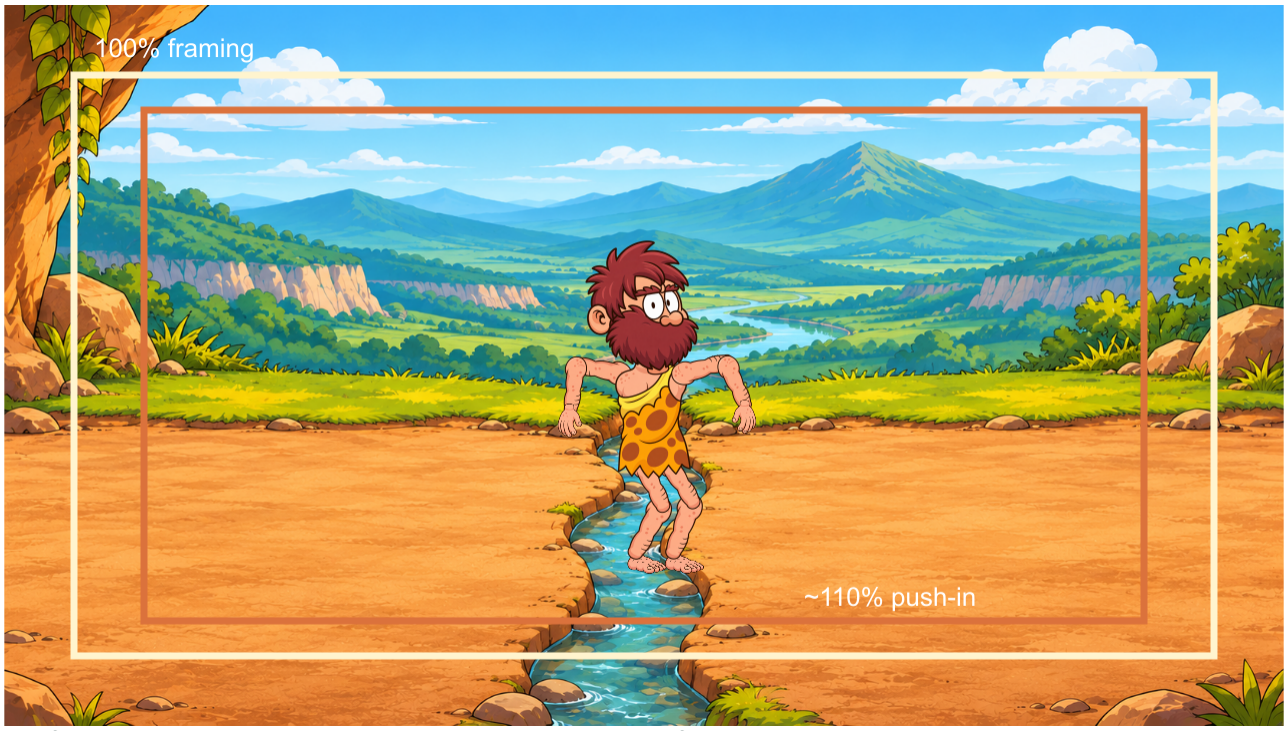
Original teacher-supplied Caveman PDF/PNG assets are preserved. New matching Caveman parts/backgrounds are AI-generated and identified in the asset register. Teaching diagrams are authored illustrations; genuine native captures are labelled separately. Supplied commercial course videos/FLAs are reference material and are not redistributed.

Publication/classroom status: see the teacher companion. This file does not certify the independent cold run, a natural voice performance, or live assessment/progress connections.

WORKBOOK 07 / 08 · MULTIMEDIA 12

Use effects and camera to tell the story

Raw-shape tweening, splash timing, sound hierarchy and a motivated push-in



Camera-framing study: a restrained push-in must retain essential story information.

Build: Add one controlled effect and one justified camera move that support the Caveman's action.

Keep: Splash test + camera comparison + mixed scene preview

Shared story: A Caveman wants a third counting stone. A stream stands in his way. He studies it, commits to a jump, retrieves the stone and returns to count his collection with enormous pride.

Document: ActionScript 3.0 · 1280 × 720 · 24 fps · three main Scenes · Graphic symbols for scrubbable character actions

Two provisional 70-minute sessions. Use the HTML guide for read-aloud, switchable supports, recovery files and narrated explanation. This print version includes the essential support prompts.

Continue the same film

Create the project in Workbook 1. For every later workbook, open the selected previous production FLA and Save As a new numbered checkpoint, preserving all three Scenes and the complete Library. Keep repair/director copies separate; return to the chosen production version.

Understand before you animate

Effects and camera should direct attention toward the story event. A visible effect is successful only if its timing and meaning fit the action.

Classic tweens transform symbol instances. Shape tweens interpolate raw vector shapes. Imported bitmap art and Graphic instances are not raw shapes. The camera belongs to the main Scene timeline for this workflow, not inside the character symbol. Sound and camera settings must be checked in the installed document type and final export.

Today’s checklist

1. Predict the result. 2. Build four stages. 3. Check in playback. 4. Refine, diagnose and direct. 5. Save and reopen.

Independent director condition

Make the landing feel heavier using pose, effect and sound. Keep the same jump distance and avoid camera shake as a substitute for impact.

What you will defend

Identify the frame that causes the splash/sound, and explain what the camera move helps the viewer notice.

Support choices

Use the step summaries, glossary, numbered checklist, different-object examples, explanation frames and optional stretch. Open the companion HTML for browser read-aloud and the narrated six-slide explanation. Supports change access to the task; the success criteria remain the same.

Your planning notes

01 · Identify the event before making the effect

SELECT: Crossing first-contact frame and water area.

TOOLS / ROUTE: Timeline review; Notes guide layer.

1. Mark the landing contact and the moment any splash should begin. If the feet land on dry bank, use a small dust effect instead of unexplained water from the feet.

2. Choose one effect: a water droplet disturbed near the stream or a bank dust puff. State what causes it.

3. Keep the first test in a separate short Graphic so the effect can be inspected without the whole film.

WHY · An effect needs a cause and location; random splashes weaken physical logic.

CHECK / SUMMARY · You can name the event and show the effect's origin beside the correct contact point.

FIX · If the effect starts before its cause, move the first visible key later. If there is no water contact, choose dust or omit the splash.

GLOSSARY · cause and effect: An event producing a visible or audible response.

DIFFERENT-OBJECT EXAMPLE · A foot on a dry path raises dust; a stone entering water creates a splash.

STRETCH · Use a tiny falling pebble as the visible cause of a water effect.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

02 · Build a genuine raw-shape tween

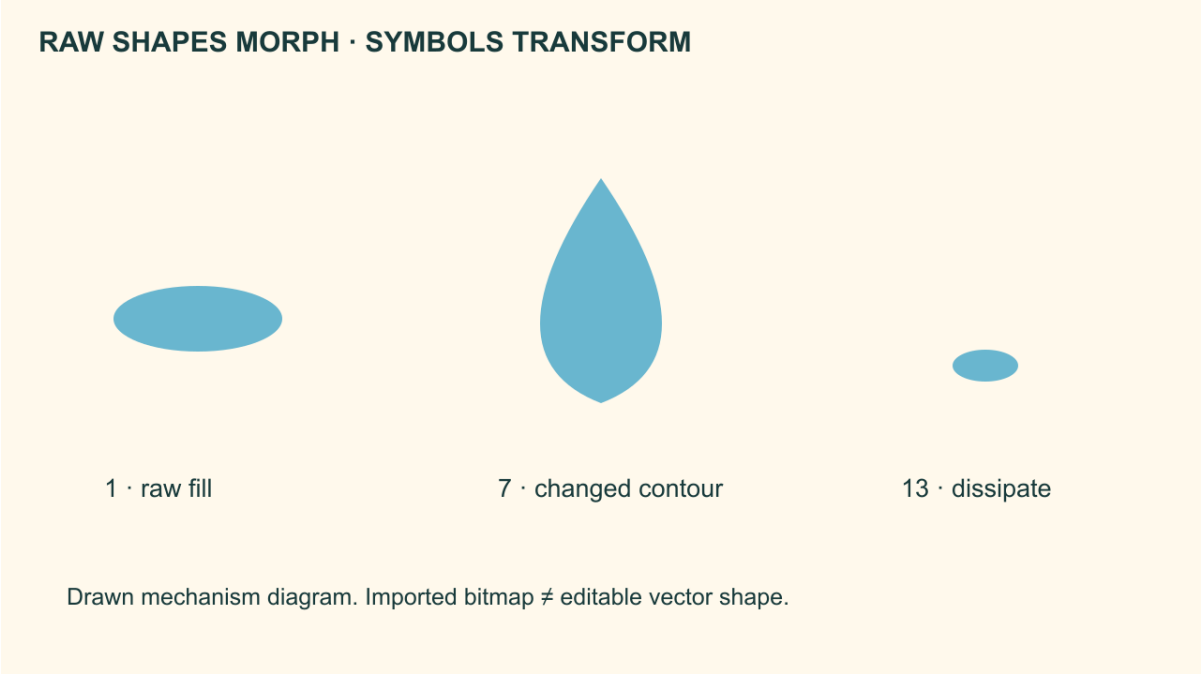
SELECT: A new Splash_Test or Dust_Test Graphic; raw drawn vector fill.

TOOLS / ROUTE: Oval/Brush tool; Insert Keyframe; right-click span → Create Shape Tween.

1. Draw a simple filled droplet or puff at frame 1 with no symbol wrapper. At frame 7 create a new key and change the raw vector shape; at 13 create a smaller dissipating shape.

2. Apply Shape Tween to the suitable spans. Scrub to check the intermediate contours. At frame 14 insert a BLANK keyframe on every visible effect layer; ordinary F5 frames would keep the last puff visible.

3. Keep the test simple: one coherent shape per layer. If you need multiple droplets, use separate layers or designed frame-by-frame drawings.



Raw-vector contour changes illustrate a shape tween; placed bitmap artwork requires a different method.

WHY · This makes the distinction between transforming a symbol and changing a raw shape visible.

CHECK / SUMMARY · The selected span is a shape tween and intermediate frames change the contour, not merely the object scale.

FIX · If the span fails, check that contents are raw vector shapes. Breaking apart a bitmap does not turn it into clean editable vector artwork.

GLOSSARY · shape tween: Interpolation between raw vector shapes.; raw shape: Editable vector fill/stroke artwork that is not a symbol instance.

DIFFERENT-OBJECT EXAMPLE · A drawn blob can morph into a teardrop; a placed photo cannot become a clean vector droplet simply by selecting Shape Tween.

STRETCH · Compare a shape-tween droplet with a frame-by-frame splash and justify the clearer one.

03 · Place the effect at its real cause

SELECT: Main Scene effect layer above the appropriate ground/water area.

TOOLS / ROUTE: Library → place effect Graphic; Play Once or timeline-limited exposure.

1. Place the effect at the marked event frame. Use Play Once for the Graphic if available and suitable, or limit its main-timeline exposure so it does not repeat.

2. Scale it to support the contact, keeping the Caveman's feet and important pose readable.

3. Scrub before, during and after contact. Confirm it is absent before the cause and clears before it distracts from retrieval.

WHY · A well-timed small effect can clarify impact more effectively than a large continuous overlay.

CHECK / SUMMARY · One caused effect appears once and clears in time for the next story beat.

FIX · If it repeats, check Graphic playback mode and parent exposure. If the contact is hidden, reduce size or move the effect behind the foot.

GLOSSARY · exposure: The frames over which artwork remains visible.

DIFFERENT-OBJECT EXAMPLE · A camera flash should appear briefly at the shutter event, not loop through the whole scene.

STRETCH · Let two small effect elements finish at different times for controlled overlap.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

04 · Add a motivated main-Scene camera move

SELECT: Me_clever main Scene; Camera tool; no nested character edit.

TOOLS / ROUTE: Camera tool → camera layer → keys and zoom controls.

1. Create a Camera layer with the Camera tool on the main Scene. Frame the three counting stones and the hand before adding movement.

2. Use a neutral camera key near the final proud phrase and a later key with a restrained push from 100 to about 110 percent. Choose an appropriate classic camera span as supported by your installed Animate version.

3. Review the full count. Keep fingers, mouth and stones within the visible camera area. The old 80–105 camera test was only a control check, not a cue plan for your new recording.

WHY · The push-in emphasizes the Caveman’s pride after the story goal is achieved.

CHECK / SUMMARY · The camera moves only when it has a purpose, and essential information remains visible throughout.

FIX · If the whole character grows but the background does not, you probably scaled the character instead of the camera. If the camera controls differ, use the static framing fallback and record the gap.

GLOSSARY · push-in: A camera move that makes the subject occupy more of the frame.; camera framing: The portion of the scene shown to the audience.

DIFFERENT-OBJECT EXAMPLE · A slow move toward a detective’s realization can focus attention on the change in thought.

STRETCH · Compare static and moving versions with a viewer and justify your selection.

05 · Build a simple sound hierarchy

SELECT: Voice and effects on separate named audio layers.

TOOLS / ROUTE: Frame sound properties; timeline preview; final movie playback.

1. Keep voice clearly intelligible. Add only approved or self-made effects for the jump/landing and optional stone contact.

2. Align impact sound with the visible contact event, allowing a deliberate stylistic offset only if you can justify it. Prevent effects from masking “one, two, three.”

3. Listen through speakers or headphones at a comfortable level and check the entire film, including Scene transitions.

WHY · Sound can establish weight and focus, but the story depends on understanding the performance.

CHECK / SUMMARY · Speech remains clear and there are no cut-off effects, unexplained repeated sounds or accidental silence at transitions.

FIX · If a sound repeats unexpectedly, inspect its Sync setting and exposure. If mixing controls are insufficient, adjust the source effect in an approved audio editor and reimport.

GLOSSARY · sound hierarchy: Prioritizing important audio, such as voice, over supporting effects.; synchronization: Aligning related picture and sound events.

DIFFERENT-OBJECT EXAMPLE · A quiet click under dialogue supports an action; a loud sustained crash can hide the words.

STRETCH · Create weight with a short low impact and a small debris tail rather than a louder peak.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

06 · Use follow-through where the viewer looks

SELECT: An actually separate hand or prop inside Count_Performance after the proud accent; the seated body/head remains one baked image.

TOOLS / ROUTE: Pose keys; staggered finishes; short playback loop.

1. Choose the separate hand or a separate stone prop to finish slightly after the whole-body accent. Work inside Count_Performance at the existing cue frame numbers. Do not move the baked head independently of its torso. Keep its motion smaller than the main accent.

2. Inspect the final silhouette and the stones while the camera is closest. Remove small movements that compete with counting.

3. Preserve a clean ending hold long enough for the result to register.

WHY · Overlap adds life when the main action remains readable.

CHECK / SUMMARY · The eye is drawn to the intended main action and the recovery decreases into rest.

FIX · If the character appears rubbery, reduce the delay and amplitude. Do not animate every piece simply because it has a layer.

GLOSSARY · overlapping action: Related parts beginning or finishing at different times.; follow-through: Motion continuing after the main action ends.

DIFFERENT-OBJECT EXAMPLE · A coat hem can settle after a person stops, while the face remains the focus.

STRETCH · Use a stone wobble as the only secondary action after the final line.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

07 · Repair an effect that hides the action

SELECT: A duplicate with an oversized effect or excessive camera zoom.

TOOLS / ROUTE: Effect scale/exposure; camera keys; before/after frames.

1. Create an obvious obstruction in the duplicate, such as a puff covering the first-contact pose.

2. Ask a viewer what happened at the landing. Repair the visual hierarchy by changing size, timing, layer order or framing.

3. Keep one before/after comparison and name the story information you recovered.

WHY · The repair evaluates effects by communication rather than spectacle.

CHECK / SUMMARY · The viewer can identify the contact and result after the repair.

FIX · If the effect still dominates, omit it temporarily and restore the essential pose first.

GLOSSARY · visual hierarchy: The order in which elements attract attention.

DIFFERENT-OBJECT EXAMPLE · An oversized title can hide the product it is meant to introduce; reducing it restores the purpose.

STRETCH · Keep the effect but reduce its contrast so the pose remains dominant.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

08 · Direct a heavier landing

SELECT: Your independent landing/effect/sound version.

TOOLS / ROUTE: Pose keys; effect timing; sound cue; static camera comparison.

1. Keep the same jump distance. Revise the landing compression, recovery spacing and supporting effect/sound to imply greater weight.

2. Compare with the original under the same static camera first. Add a camera choice only if it helps read the action.

3. Save and reopen. Explain the cause of each effect and which change most altered the impression of weight.

WHY · The changed condition tests animation principles across pose, spacing and supporting media.

CHECK / SUMMARY · The heavier version remains readable and physically motivated, with clear speech later in the film.

FIX · If the difference depends only on louder sound, return to contact, compression and recovery.

GLOSSARY · weight cue: A visible or audible choice suggesting mass and force.

DIFFERENT-OBJECT EXAMPLE · Independent task: no heavy-landing solution is supplied.

STRETCH · Make a lighter landing as a third comparison without changing the character's size.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

Keep a small, useful evidence record

At frame ___, I chose ___ so the viewer could ___.
The problem was caused by ___. I changed ___ and checked ___.
Under the changed condition, I revised ___ because ___.

One repair

One independent decision and its trade-off

Delivery check

The selected file reopens. The result plays as intended. Joints/contact and audio were checked where relevant. The changed condition is solved. Required production files and a brief defence are retained.

Method references and attribution

A1 · Adobe Animate: symbols and instances · Adobe reference

A2 · Adobe Animate: timeline layers · Adobe reference

A3 · Adobe Animate: shape tweening · Adobe reference

A4 · Adobe Animate: parenting view · Adobe reference

A5 · Adobe Animate: sound · Adobe reference

A6 · Adobe Animate: camera · Adobe reference

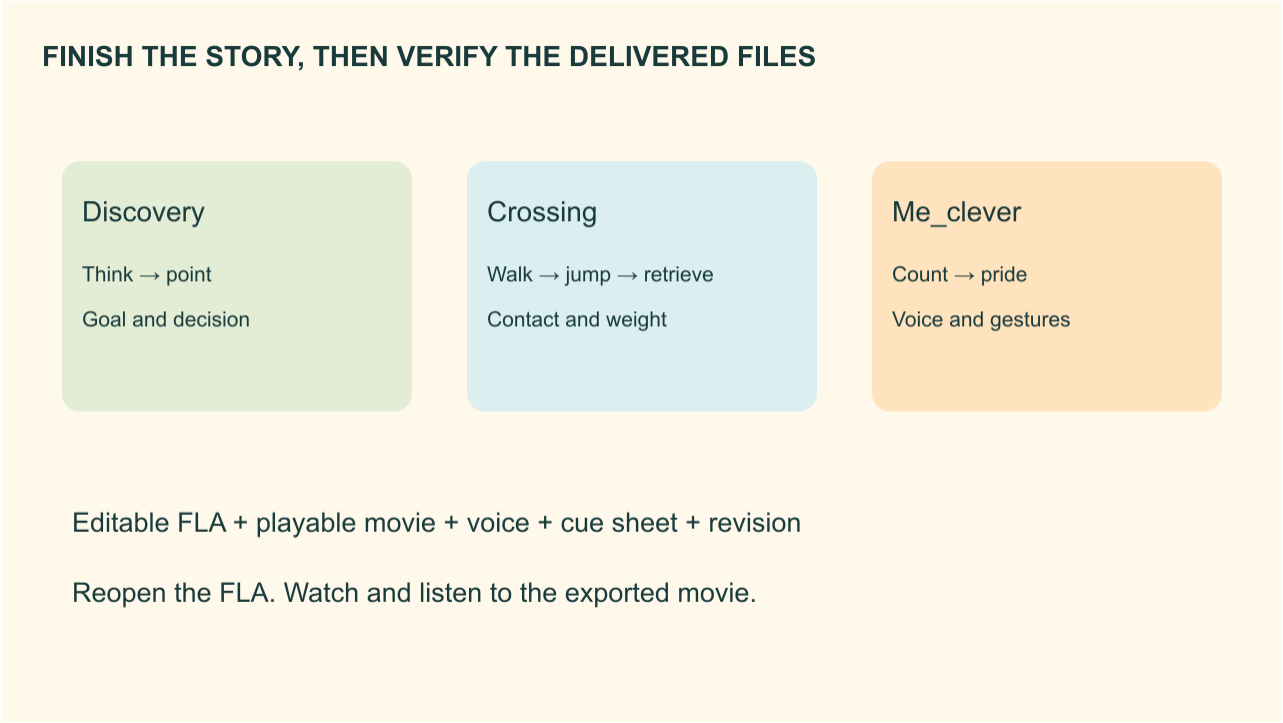
Original teacher-supplied Caveman PDF/PNG assets are preserved. New matching Caveman parts/backgrounds are AI-generated and identified in the asset register. Teaching diagrams are authored illustrations; genuine native captures are labelled separately. Supplied commercial course videos/FLAs are reference material and are not redistributed.

Publication/classroom status: see the teacher companion. This file does not certify the independent cold run, a natural voice performance, or live assessment/progress connections.

WORKBOOK 08 / 08 · MULTIMEDIA 12

Finish a complete Caveman film

Scene continuity, critique, revision, independent direction and delivery



Final assembly map links selected actions to one complete film and its verified delivery files.

Build: Deliver a coherent, playable film and an editable project with evidence of a purposeful revision.

Keep: Final FLA + playable movie + source audio + cue sheet + concise production evidence

Shared story: A Caveman wants a third counting stone. A stream stands in his way. He studies it, commits to a jump, retrieves the stone and returns to count his collection with enormous pride.

Document: ActionScript 3.0 · 1280 × 720 · 24 fps · three main Scenes · Graphic symbols for scrubbable character actions

Two provisional 70-minute sessions. Use the HTML guide for read-aloud, switchable supports, recovery files and narrated explanation. This print version includes the essential support prompts.

Continue the same film

Create the project in Workbook 1. For every later workbook, open the selected previous production FLA and Save As a new numbered checkpoint, preserving all three Scenes and the complete Library. Keep repair/director copies separate; return to the chosen production version.

Understand before you animate

A finished film coordinates story, performance, motion, sound and editing. Completion includes opening the deliverables and checking what the audience actually receives.

The working FLA preserves editability; the rendered movie carries the viewing experience. An editor timeline or successful export command does not prove that the movie plays correctly. Native author tests are separate from a student cold run and from school assessment registration. This package does not claim live progress synchronization.

Today’s checklist

1. Predict the result. 2. Build four stages. 3. Check in playback. 4. Refine, diagnose and direct. 5. Save and reopen.

Independent director condition

Choose one director condition from the previous workbooks and carry it consistently through the final film. Keep the shared story spine and explain the resulting choices.

What you will defend

Show a before/after revision, identify the principle involved, and demonstrate that your final editable and playable files agree.

Support choices

Use the step summaries, glossary, numbered checklist, different-object examples, explanation frames and optional stretch. Open the companion HTML for browser read-aloud and the narrated six-slide explanation. Supports change access to the task; the success criteria remain the same.

Your planning notes

01 · Assemble the selected actions in story order

SELECT: Main Scenes Discovery, Crossing and Me_clever.

TOOLS / ROUTE: Scene panel; Library action instances; main timeline.

1. Use this worked assembly map: Crossing 1–72 = Walk_24 looping for three cycles; 73–84 = a 12-frame Stop; 85–120 = Jump_36; 121–156 = a 36-frame Pickup; 157–192 = a result hold or optional far-bank carrying departure. These authored values can be revised after the animatic check. Discovery uses your selected Think_Point; Me_clever uses the measured recording.

2. Insert a new Character main-timeline key at every action start and Swap to that action Graphic. Set Jump_36, Stop and Pickup to Play Once/First 1 with exposure exactly covering their local action frames; use Single Frame on the chosen last pose for a hold. Walk_24 alone loops. Main frame = action start + local frame − 1; Jump local 28 therefore lands at Crossing 112.

3. The jump's travel is inside Jump_36. Its outer instance stays at the take-off anchor; at Pickup frame 121 place the next action at the landed root position, not back at the take-off anchor. Match the last/first body pose at each hand-off. Remove temporary references, then test the entire movie and check no action loops unexpectedly.

WHY · The final film needs intentional hand-offs between actions, not merely completed isolated tests.

CHECK / SUMMARY · Goal → decision → crossing → retrieval → counting reads as one story.

FIX · If an action starts on the wrong internal frame, inspect its Graphic playback and First setting and the parent keyframe where it is placed.

GLOSSARY · assembly: Combining selected actions into the final sequence.; hand-off: The transition from one action or shot to the next.

DIFFERENT-OBJECT EXAMPLE · Edited interview clips need meaningful transitions even when each clip is good individually.

STRETCH · Try a brief reaction hold between retrieval and the cut back to the cave.

02 · Check continuity with sound muted

SELECT: Full movie preview, sound temporarily muted in playback.

TOOLS / ROUTE: Test Movie; scene order; frame-by-frame inspection.

1. Watch for screen direction, prop ownership, foot contact, character scale and background ground height across transitions.

2. Confirm that the time-ellipsis cut back to the cave is understandable. The third stone must be visibly acquired before it appears in the collection.

3. Note one exact frame or cut that causes uncertainty and repair it in the FLA. Restore sound for later checks.

WHY · A coherent silent action makes the spoken ending more satisfying and reveals visual dependence on explanation.

CHECK / SUMMARY · A new viewer can recount what happened without you narrating over the film.

FIX · If the return cut is confusing, add a small departure/arrival cue instead of a lengthy explanatory title.

GLOSSARY · screen direction: The apparent left/right direction of movement or attention.; continuity check: A review of consistent relationships across frames and shots.

DIFFERENT-OBJECT EXAMPLE · A character leaving right and entering right may imply a reversal unless the edit explains the geography.

STRETCH · Make a contact sheet of the last and first frames around every cut.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

03 · Review principles through visible evidence

SELECT: The final action keys and cue sheet.

TOOLS / ROUTE: Timeline and short playback loops; evidence notes.

1. Identify one staging choice, one designed breakdown, one timing/spacing choice and one anticipation/landing recovery.

2. Show one lip-sync cue and one true hand Swap. Explain what each contributes to this story rather than reciting a definition.

3. Choose only the clearest evidence. Four precise frame references are stronger than a page of principle names with no visible connection.

WHY · Assessment should reflect decisions and understanding demonstrated in the production.

CHECK / SUMMARY · Your explanations point to identifiable frames, poses or audio events in your own file.

FIX · If a named principle has no visible example, either revise the action or remove the claim.

GLOSSARY · production evidence: Specific work showing a skill or decision in context.

DIFFERENT-OBJECT EXAMPLE · A designer can defend contrast by pointing to a readable label under a tested condition.

STRETCH · Identify a principle you deliberately used very little and explain why.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

04 · Run a focused audience review

SELECT: A playable draft and a viewer who has not followed your timeline.

TOOLS / ROUTE: Playback; three neutral review prompts.

1. Ask: What does he want? Where did the movement feel unclear or weightless? Did the ending and spoken count agree?

2. Let the viewer watch without interruption. Record one observation in their words, without collecting personal details.

3. Choose one high-impact revision. Save a before copy, revise, and replay the same segment.

WHY · Targeted feedback reveals communication problems that familiarity can hide.

CHECK / SUMMARY · The revision responds to observed audience difficulty, not only personal preference.

FIX · If feedback is vague, ask which moment caused it; do not tell the viewer what they were supposed to think.

GLOSSARY · critique: Specific observation used to improve a work.; iteration: A cycle of making, testing and revising.

DIFFERENT-OBJECT EXAMPLE · A user saying “I missed the button” suggests a visibility test, not a total redesign.

STRETCH · Use a second viewer to check whether the repair solved the original problem.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

05 · Finish the director variation

SELECT: A final duplicate with one selected changed condition.

TOOLS / ROUTE: Save As → Caveman_final_director.fla; consistent scene edits.

1. Select one earlier condition: harder-to-see goal, a post-pickup carrying departure, wider stream, changed vocal reading or heavier landing.

2. Carry that decision through all affected Scenes. A wider jump may require revised staging and retrieval timing; a new recording requires a new cue sheet.

3. Write a short defence: condition → production choices → visible evidence → remaining trade-off.

WHY · Transfer is shown by coordinating consequences, not adding an unrelated extra effect.

CHECK / SUMMARY · The independent version preserves the story spine and solves the stated condition consistently.

FIX · If the change exists only in an isolated test, replace the corresponding final action and recheck the adjacent hand-offs.

GLOSSARY · coherence: Parts working together toward a consistent purpose.; trade-off: An improvement that requires a cost or compromise elsewhere.

DIFFERENT-OBJECT EXAMPLE · Independent task: use the story, contact and synchronization criteria to judge your own solution.

STRETCH · Compare the director version with the guided film and justify your final selection.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

06 · Export a playable movie

SELECT: Saved final FLA; entire movie rather than current isolated symbol.

TOOLS / ROUTE: File → Export → Export Video/Media; choose the installed output route.

1. Save the FLA. Use Export Video/Media for the full movie and select 1280 × 720 with the project's 24 fps. Inspect the available settings in the installed Animate/Media Encoder workflow.

2. If Adobe Media Encoder is available, complete the render to a standard playable movie such as MP4/H.264. If that route is unavailable, retain the intermediate or PNG sequence plus WAV, assemble them at 24 fps in an approved editor and export the playable movie there. Keep the source materials; the sequence alone is not the required viewing copy. If no approved render route is available, mark the viewing copy incomplete rather than claiming final delivery.

3. Do not submit a SWF as the only viewing copy. Name the final playable file clearly and keep it with its source project.

WHY · A viewing copy must play outside the authoring timeline with sound and all Scenes.

CHECK / SUMMARY · The exported file exists, opens in a separate player and includes the full ending.

FIX · If only one Scene exports, check the selected range and whether you were inside a symbol. If audio is missing, inspect source sound and the chosen export route.

GLOSSARY · render: Producing a viewing file from the editable project.; codec: A method used to encode/decode video or audio.

DIFFERENT-OBJECT EXAMPLE · A slideshow's editable file and its exported PDF serve different purposes; both must be checked.

STRETCH · Compare a smaller viewing copy with the master for readable mouth and hand details.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

07 · Reopen and verify the delivery

SELECT: The exported movie in a separate player and the saved FLA.

TOOLS / ROUTE: Open movie; File → Open FLA; full playback.

1. Play the movie from start to finish with sound. Check cuts, contact, lip sync, fingers, camera crop and the final hold.

2. Close and reopen the FLA. Verify Scene names, Library actions and the saved cue timing. Confirm linked/source audio remains available.

3. Keep one delivery checklist recording the filenames, actual runtime and any limitation. A successful save or export message alone is not proof.

WHY · The delivered files, not the workspace state, are what the audience and teacher can use.

CHECK / SUMMARY · Both final files open and agree on the selected version, duration and ending.

FIX · If the movie and FLA differ, render again from the saved chosen version and replace the mislabeled viewing copy deliberately.

GLOSSARY · verification: Checking the actual delivered result against its intended behavior.

DIFFERENT-OBJECT EXAMPLE · A website screenshot cannot prove a download works; the downloaded file must be opened.

STRETCH · Check the movie on a second available player or classroom device.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

08 · Hand in a concise production package

SELECT: Final project folder; your teacher's named collection route.

TOOLS / ROUTE: Folder organization; upload only to the stated classroom destination.

1. Include the final FLA, playable movie, original voice recording, cue sheet, six-panel storyboard and one before/after revision explanation.

2. Keep the reflection short: one strong decision, one solved problem and one next improvement. Cite any third-party assets used; identify supplied AI-generated Caveman artwork as such.

3. Use the teacher's actual collection instructions. This guide can save/download a local evidence copy; it does not register assessment IDs, submit work or prove shared app progress.

WHY · A clear package lets someone review the product, process and understanding without guessing which file is final.

CHECK / SUMMARY · The package has one clearly selected final version and enough evidence to demonstrate the taught skills.

FIX · If the school submission route is not named, keep the complete local package and obtain that route from the teacher before sending it anywhere.

GLOSSARY · attribution: Identifying the origin and creator/type of an asset.; delivery package: The selected final files and necessary supporting evidence.

DIFFERENT-OBJECT EXAMPLE · Independent evidence: explain your own choices, not a supplied model answer.

STRETCH · Prepare a 60-second oral defence using your own timeline as evidence.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

Keep a small, useful evidence record

At frame ___, I chose ___ so the viewer could ___.
The problem was caused by ___. I changed ___ and checked ___.
Under the changed condition, I revised ___ because ___.

One repair

One independent decision and its trade-off

Delivery check

The selected file reopens. The result plays as intended. Joints/contact and audio were checked where relevant. The changed condition is solved. Required production files and a brief defence are retained.

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